International Journal of Thermofluids 20 (2023) 100486

Contents lists available at ScienceDirect

International Journal of Thermofluids

journal homepage: www.elsevier.com/locate/ijtf

Significance of a permeable semi-circular body and magnetic field on the double-diffusive traits

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ARTICLE INFO

Keywords:
Semi-circular obstacle
Double-diffusion
Magnetic effect
LBM
Permeable structure
Unsteady

ABSTRACT

This study delves into the occurrence of double-diffusive natural convection while being affected by a magnetic field within a cold square enclosure, containing a heated and concentrated semi-circular body made permeable. The fusion of both flat and curved boundaries, combined with the permeable attributes, stands as the central impetus driving this inquiry. The main aim is to scrutinize the impact of buoyancy force, permeability, magnetic field strength, and angle on the double-diffusive convection taking place inside and around the permeable structure. To achieve this, a comprehensive parametric investigation is undertaken, emphasizing the importance of Darcy number (Da), buoyancy ratio (BR), Hartmann number (Ha), and magnetic field angle (γ_M) in relation to the flow, heat, and concentration transfer. We utilize the Lattice Boltzmann method, incorporating the Darcy-Brinkmann-Forchheimer equation as a source term, to precisely address the complex flow within the porous medium. The results will provide insights into the importance of a semi-circular body with porous characteristics in the context of thermo-solutal convection influenced by magnetic effects. These findings hold practical significance in various applications, including electronic cooling, pollutant removal, and food processing industries. The inquiry discloses that elevating the buoyancy force, permeability, and the angle of the magnetic field leads to advantageous heat and concentration characteristics. Furthermore, positioning the magnetic force in opposition to gravity reduces the restraining impact of the magnetic force on the mean Nusselt number (Nu_M) and the mean Sherwood number (Sh_M). Remarkably, instances of unsteady behaviour are predominantly noted when the BR takes on a negative value, and these phenomena's intensity and frequency are heightened by the Da and the γ_M. Remarkably, there are a few irregular spikes detected in the Nu_M and Sh_M, which can be attributed to the emergence of a strong singular vortex in the upper part of the enclosure.

1. Introduction

Thermo-solutal convection, an interesting phenomenon stemming from the interplay between fluids characterized by disparate temperature and concentration gradients, has brought about transformative implications across an array of real-world applications. Its influence spans across a diverse spectrum of fields, encompassing disciplines like oceanography, environmental surveillance, geothermal energy exploitation, materials science, astrobiology investigations, climate modelling endeavours, and the dynamic landscape of the food and beverage industry. Furthermore, the domain of double-diffusion extends its influence into the realm of porous media, unveiling remarkable utilities in areas like enhanced oil recovery techniques, geologic carbon sequestration strategies, advancements in heat transfer within building materials, and even in the biology and biomedicine. In the realm of battery systems, a profound comprehension of double diffusion occurring within porous electrodes has emerged as a pivotal factor in the conceptualization and refinement of battery technologies. By understanding the transportation of ions and solvents through these porous structures, scientists can achieve higher performance and efficiency not only in lithium-ion batteries but also in other energy storage innovations. The focus directed towards porous structures in the context of heat transfer applications is rooted in their inherent capacity to elevate convective heat transfer, heighten thermal insulation, facilitate heat accumulation and restitution, function as highly effective heat dissipation mediums, and catalyze boiling and condensation processes. These advantages propel innovation across diverse industries, contributing significantly

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https://doi.org/10.1016/j.ijft.2023.100486
Received 3 September 2023; Received in revised form 11 October 2023; Accepted 11 October 2023
Available online 17 October 2023
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Nomenclature

Notations

B	Magnitude of magnetic field
c_F	Non-dimensional Forchheimer term
C_i^{eq}	Equilibrium distribution function of concentration in direction i
C_p	Specific heat at constant pressure, [J/kg K]
Da	Darcy number, K/S^2
e_i	Discrete lattice velocity in direction i, $\frac{\Delta x_i}{\Delta t}$
f_i	Particle density distribution function in direction i
F_b	Boussinesq force term, [N]
F_i	Total force term due to porous medium
Ha	Hartmann number
Le	Lewis number
K	Permeability of the material, [m^2]
g_i	Temperature distribution function in direction i
N	Number of lattices on the enclosure
Pr	Prandtl number, $\frac{\nu}{\alpha}$
S	Diameter of the semi-circular body
S_M	Entropy generation due to magnetic field
S_χ	Entropy generation due to concentration transfer
Sh_M	Mean Sherwood number, $(\partial\chi/\partial n)$
v	Non-dimensional y-component velocity
V	Auxiliary velocity, [m s^{-1}]
x, y	Horizontal vertical coordinates
BR	Buoyancy ratio, $(\beta_C \Delta C/\beta_T \Delta T)$
C_i	Concentration distribution function in direction i
c_s	Speed of sound, [m s^{-1}]
D	Mass diffusivity, [m^2 s^{-1}]
d_{pp}	Particle diameter of porous medium, [m]
F	Body force due to the presence of the porous medium
f_i^{eq}	Equilibrium distribution function of density in direction i
F_M	Magnetic force
g	Gravitational acceleration, [m s^{-2}]
L	Enclosure height
k	Thermal conductivity, [W/mK]
Ma	Mach number, (u/c_s)
g_i^{eq}	Equilibrium distribution function of temperature in direction i
Nu_M	Mean Nusselt number, $(\partial\theta/\partial n)$
Ra	Rayleigh number, $\frac{g\beta\Delta T L^3}{\alpha\nu}$
S_T	Total entropy generation
S_θ	Entropy generation due to heat transfer
S_ψ	Entropy generation due to fluid friction
u	Non-dimensional x-component velocity
U	Actual velocity of porous medium, [m s^{-1}]
w_i	Weighing factor in direction i

Greek symbols

ρ	Fluid density, [kg m^{-3}]
τ'	Dimensionless relaxation time for temperature
μ	Dynamic viscosity of fluid [N s m^{-2}]
Δt	Time step, [s]
θ	Dimensionless temperature, $((T - T_c)/(T_h - T_c))$
ϵ	Porosity
β_C	Concentration expansion coefficient
σ	Electrical conductivity, [s/m]
γ_M	Magnetic field angle
τ	Dimensionless relaxation time for density
τ''	Dimensionless relaxation time for concentration
Δx	Lattice space
ν	Fluid kinematic viscosity, [m^2 s^{-1}]
χ	Dimensionless concentration, $((C - C_c)/(C_h - C_c))$
β_T	Thermal expansion coefficient, [°C^{-1}]
α	Thermal diffusivity, [m^2 s^{-1}]
τ_t	Non-dimensional time

Subscripts

i	Lattice link direction
c	Low
w	Wall
h	High

to energy efficiency, thermal regulation, and the advancement of sophisticated heat transfer systems. Agglomerated solid structures can indeed exhibit porous attributes, finding pertinent applications within heat transfer procedures. In such instances, the spatial configuration of these clusters plays a pivotal role in dictating heat dispersal and retention dynamics. The captivating interplay between buoyant forces stemming from thermal and concentration gradients, in conjunction with the influence of magnetic fields and the intricate dynamics of porous formations, continues to engross engineers, motivating them to unearth novel aspects and exploit the complete potential of the double-diffusive phenomenon. To unlock the entirety of the potential inherent in the ongoing exploration, a profound comprehension of the invaluable insights drawn from both well-established and pioneering studies becomes indispensable. Therefore, a discussion of several pivotal studies follows in the subsequent section.

An endeavour by Sezai and Mohamad [1] aimed to explore the characteristics of buoyancy-driven fluid motion influenced by temperature and solute concentration gradients in a cube-shaped enclosure. The investigation highlighted that the mean Sherwood (Sh_M) and Nusselt (Nu_M) numbers from two-dimensional analyses exhibited a noticeable elevation in contrast to their three-dimensional counterparts. This disparity was especially pronounced within the range of buoyancy ratio (BR) spanning from −1.3 to +1.3. The alterations in heat and mass transfer within a vertical annular space, examined up to elevated thermal Rayleigh numbers (Ra_t) by Chen et al. [2], exposed a consistent upward trajectory in the values of both Nu_M and Sh_M for the heated and concentrated boundary as the proportions of the enclosure grew. The investigation into double-diffusive combined natural convection by Kuznetsov and Sheremet [3] pointed out that the dimensions of the concentration source intensify the dispersion of heat in the cubical cavity filled with air. Additionally, they noted transient behaviours in the transfer of heat and mass as the size of the contaminant source was augmented. The inquiry into the interplay of concentration and thermal sources on the double-diffusive behaviours within a rectangular cavity, as conducted by Nikbakhti and Rahimi [4],

revealed that situating the source at the bottom led to the most pronounced thermo-solutal features. Lately, a multitude of investigations have delved into the intricacies of double-diffusive behaviours, considering shifts in the working fluid and/or the shape of the enclosure. In this context, Aghighi et al. [5] delved into the thermo-solutal properties of a non-Newtonian Casson fluid confined within a square cavity. They analysed influential parameters like the Lewis number (Le), buoyancy ratio (BR), yield number (Y), and Rayleigh number (Ra). Their findings demonstrated that an increase in Le curbs the impact of buoyant forces on the hydro-thermal characteristics. In recent times, Azad et al. [6] examined the influence of dimensionless time and BR on the thermo-solutal mixed convection behaviours within a square cavity featuring dual ventilations. Their investigation unveiled that, under a consistent dimensionless time, the concentration and thermal gradients were observed to remain steady. Jain et al. [7] investigated the impact of a nanofluid-filled double-wavy enclosure on natural convection driven by thermal and solute gradients. In addition to conventional factors like Le, BR, Ra, and power-law index, they also examined the changes in Brownian motion and thermophoresis parameters in their research. Their findings clarified that these two parameters negatively influenced thermal dissipation, yet yielded positive effects on mass transfer. Plenty of investigations have been carried out on the subject of natural, thermo-solutal convection with different fluids [8–10]. Lately, Tayebi et al. [11] scrutinized the effects of Nano-Encapsulated Phase Change Material (NEPCM) within enclosures characterized by I-shaped and wavy walls, focusing on alterations in double-diffusive phenomena. Their investigation unveiled that a reduction in the latent heat of the NEPCM, coupled with an elevation in its fusion temperature, resulted in a substantial enhancement of mass transfer.

Leveraging the potential of magnetic field offers a captivating avenue for governing and manoeuvring double-diffusive free convection, resulting in a notable capability to manage the thermal and mass transfer traits.Magneto-hydrodynamics (MHD) is a scientific discipline that focuses on the investigation of the dynamic interplay between magnetic fields and convective motion. On this subject wide spectrum of investigations have been carried out [12–15]. Chamkha and Al-Naser [16] examined the repercussions of buoyancy-driven double-diffusive flow within a rectangular enclosure, where consistent mass and heat fluxes, along with divergent gradients of concentration and temperature, were subjected to the influence of magnetic strength. Their discoveries unveiled a decline in thermal dissipation and fluid movement within the enclosure as a result of the magnetic field. In a subsequent investigation conducted by the same researchers [17], the original fluxes were substituted with consistently maintained concentration and temperature on the sidewalls. The emphasis of this study was on the characteristics of buoyancy-driven double-diffusive flow. The study revealed that the dynamic behaviour of thermal and concentration transfers could be proficiently managed through the application of a magnetic force. The exploration carried out by Teamah [18] within a rectangular enclosure, encompassing varying intervals of Ra_t, Hartmann number (Ha), and BR, illustrates that an escalation in Ha significantly hinders the magnitudes Nu_M and Sh_M. Moreover, the investigation into the influence of an inclined magnetic force and cavity on thermo-solutal free-convection, undertaken by Teamah et al. [19], disclosed that the highest levels of concentration and heat transfer rates were attained at an inclination angle of 150°, encompassing all the examined combinations of Ha and Ra_t. On the contrary, Sathiyamoorthi and Anbalagan [20] emphasized that the angle leading to the maximum rates of heat and mass transfer is dependent on the magnetic field, particularly noticeable at higher levels of Ra_t. The study performed by Maatki et al. [21] within a cubic cavity, subject to differential solute concentration and heating, delved into the importance of the angle of the magnetic field. Their findings pointed out that the inclination exerts an adverse influence on thermal convection. It is crucial to recognize that the influence of the magnetic force and its angle also relies on the direction of fluid movement, which is subsequently influenced by the conditions set at the solute and thermal boundaries. Kushawaha et al. [22] conducted an investigation into magnetohydrodynamics (MHD) of buoyancy-driven thermo-solutal flow within a square enclosure, featuring a left wall with exponential heating and concentration. Their study focused on impact of magnetism on both magnetic and non-magnetic nanofluids. Their findings indicated that the magnetic nanofluid yielded positive results even at lower values of Ra when influenced by a magnetic field. Moreover, an increase in the magnetic angle was found to enhance the rates of heat and concentration transfer. Moreover, recent research works [23–26] involving enclosures of various shapes with a centrally heated and concentrated circular source also underscores that the coefficients governing solute and heat flux diminish as the Ha increases. Parveen et al. [27] delved into the irreversibilities within an inclined dome-shaped cavity featuring discrete heating. Their investigation highlighted that the irreversibilities tend to diminish as the Ha and the orientation of the magnetic force increase. Furthermore, they noted that such a dome-shaped enclosure promotes streamlined fluid flow, consequently resulting in a substantial augmentation of heat and mass transfer rates. Chuhan et al. [28] embarked on an exploration concerning the optimization of irreversibilities within a wavy enclosure filled with non-Newtonian fluid under the influence of magnetohydrodynamics (MHD) and double-diffusive buoyancy-driven flow. The objective was to pinpoint conditions that would minimize entropy generation and enhance overall system efficiency. The investigation unveiled notable fluid friction irreversibilities present at the wavy wall of the enclosure.

Incorporating a porous medium introduces substantial alterations in hydro-thermal and mass transfer behaviours. These changes arise from factors such as increased hydraulic resistance, expanded surface area, and the arrangement of pores within the medium. Furthermore, the thermal conductivity of the porous material plays a significant role in shaping these modifications. The integration of these complex mechanisms finds practical applications in various scenarios, as discussed earlier in this section. Mohamad and Bennacer [29] conducted an examination of buoyancy-driven thermo-solutal flow within a cavity containing a saturated porous structure. Through the application of differential heating and a vertical concentration gradient, the study unveiled that augmenting Le notably heightens the Sh_M, while the Nu_M shows a modest increase. Moreover, the study observed the occurrence of unstable flow patterns when the Le assumes a large value (i.e. $Le = 100$), in conjunction with a negative BR exceeding unity. Mohamad et al. [30] investigated the characteristics of thermo-solutal convection within a cubic enclosure filled with a porous medium, wherein the left wall exhibits elevated concentration and temperature. The study underscored that when Da attains higher levels, an asymmetric flow scenario emerges, highlighting the need for three-dimensional analysis to comprehensively comprehend the flow structure. Sankar et al. [31] directed their attention towards double-diffusive convection featuring discrete solute and temperature sources within a vertical porous annular space. The study illuminated that the pivotal threshold of BR, which results in flow transition and reversal, is markedly influenced by parameters such as Ra_t, Da, Le, and the placement of the sources within the system. The attributes of MHD natural convection and associated irreversibilities in an enclosure filled with differentially heated $Ag - H_2O$ nanofluid and porous material were scrutinized by Fares et al. [32]. The investigation unveiled that the Da exhibits a tendency to generate a higher amount of irreversibilities within the system. In the research conducted by Wang et al. [33], which investigated the effects of $Al_2O_3 - H_2O$ nanofluid and a square enclosure filled with porous material on thermo-solutal natural convection, the findings indicated that the inclination of the enclosure had negligible impact at Da values. However, its influence became significant as Da values increased. Hadidi et al. [34] investigated the phenomenon of thermo-solutal free-convection within an enclosure while accounting for different thermal conductivity and thickness values of a porous layer. Their outcomes revealed that increases in the stream function value were notable only when the thickness of the

porous layer was 0.2 or 0.5. Under these conditions, the rates of heat and mass transfer experienced a decrease. However, extending the layer thickness beyond these values impeded the kinetic energy of the fluid. In contemporary research, a number of investigations [35–42] have underscored the significance of porous media in comprehending the mechanisms of heat and mass transfer, alongside the influence of MHD on these phenomena. These studies illuminate the intricate interplays involved and offer valuable perspectives for enhancing heat and solute transfers in diverse practical applications.

Researchers across various engineering disciplines have fervently immersed themselves in the intricate domain of temperature and concentration dynamics stemming from internal heat and solute sources. This captivating field of study has engrossed their steadfast focus due to its profound relevance across numerous domains. From the sphere of environmental management, enabling air quality control and pollutant eradication, to the captivating realms of nano-electronic device fabrication, cooling for nuclear reactors, and advancements in catalytic converter technology, the repercussions of these explorations resonate extensively. A sequence of investigations was carried out to comprehend the effects of a heated and concentrated source within a ventilated cavity [43], the position of the source [44], and the location of vent openings [45] on thermo-solutal results, involving a cylindrical body situated within a ventilated enclosure. Xu et al. [43] noted that as the BR value increases within the existing forced convection regime, there is a reduction in heat transfer, except when $Le = 0.1$. Furthermore, when the cylindrical body is positioned in the lower region, the effect of BR on the Sh_M at a Richardson number (Ri) of 0.01 is determined to be negligible [44]. Evidently, the impact of BR on thermo-solutal convection becomes notably pronounced within the free-convection regime. However, as outlined by Xu et al. [45], placing the cylindrical body in the upper-middle region yields favourable mass transfer effects, particularly when the BR is equal to unity. The study conducted by Nazari et al. [46] investigated how various factors affecting double-diffusion, namely Ra, BR, aspect ratio, and Le, impact the movement of heat and solutes around a square obstacle placed inside a confined space. The research highlighted that increasing the buoyancy ratio from a positive value resulted in higher values for Nu_M and Sh_M. Conversely, reducing the negative BR led to a decrease in the rates of heat and mass transfer. In a computational investigation carried out by Kefayati and Tang [47], the impacts of parameters such as Ra, Le, BR, the radius of a cold cylinder, and its positioning were explored in relation to double-diffusion phenomena within a cavity filled with Carreau fluid. The research findings demonstrated that shifting the cylinder's vertical position from below to above the cavity caused a notable reduction in the Nu_M and Sh_M. Furthermore, positioning the cylinder horizontally away from the cavity's centre resulted in an enhancement of the rates at which heat and mass were transferred. Building upon the aforementioned concern, Kefayati and Tang [48] delved deeper into the changes in entropy generation within the enclosure. The outcomes of their investigation revealed that increasing the dimensions of the cylinder resulted in a greater level of irreversibilities. Furthermore, situating the cylinder in closer proximity to the bottom side led to a decreased entropy value in comparison to other placements. Performing a comprehensive assessment through a second-law analysis within a cavity containing a porous medium and including internal sources of heat and solute, Siavashi et al. [49] conducted an extensive exploration. The research findings unveiled that the existence of two separate internal sources within the enclosure notably boosts the rates of both concentration and heat transfer. Furthermore, it was observed that the presence of the lowest entropy levels coincided with enhanced values of Nu_M and Sh_M when two vertical sources were strategically placed within the enclosure. The influence of vertically oriented heat-conducting partitions on the thermal and solutal results within a cavity was scrutinized by He et al. [50]. The research exposed that the thermal conductivity ratio of the partition plays a pivotal role in shaping the thermal and concentration distributions, especially when the partitions are placed near the heated and concentrated sidewall. In a contemporary investigation carried out by Rashid et al. [51], the emphasis was placed on the analysis of a crown-shaped enclosure with a solid circular cylinder at its centre, subjected to heating. The research considered the presence of a magnetic field in combination with the use of Casson fluid. The findings of the study illustrated that augmenting the Casson parameter resulted in an increase in the heat transfer rate. Conversely, the mass transfer rate exhibited a reduction as a consequence of elevating this parameter.

Extensive inquiries have been conducted regarding the convective characteristics of semi-circular cylinders due to their substantial importance across various contexts such as compact heat exchangers, food thermal processing, and polymer shaping. Several noteworthy studies [52–56] have made significant contributions to this domain. In a particular investigation, Chandra and Chhabra [52] scrutinized the characteristics of buoyancy-driven phenomena around a semi-circular cylinder. The study emphasized the importance of parameters like the power-law index (n), Grashof number (Gr), and Prandtl number (Pr). The researchers highlighted the highest Nusselt number occurring at the corners of the cylinder, with an increase that correlated with higher values of Gr. In a separate inquiry, Mahajan et al. [54] examined the influence of an inclined and heated semi-circular barrier situated in a square enclosure on the nature of natural convection. Intriguingly, the highest heat transfer was observed when the cylinder was positioned at a horizontal angle of 90°, aligning its edge with the right-side wall of the enclosure. This outcome was particularly significant under circumstances characterized by elevated Ra and Pr values. The fascinating process of heat transfer among clustered bodies entails the sharing of thermal energy between multiple bodies that are closely positioned or directly connected. This captivating arrangement gives rise to unique heat transfer dynamics, akin to the flow patterns witnessed within porous materials. In scenarios like compact heat exchangers, packed bed reactors, biological tissues, and packed bed filters, the setup of densely grouped entities mimics the traits of a porous medium, where the movement of heat and fluids occurs through interconnected pathways. Considerable scholarly effort has been directed towards delving into this topic in recent years, encompassing a range of convection scenarios and a diverse array of shapes for porous structures [57–62]. Within the domain of natural convection, a consistent observation comes to light, revealing a marked elevation in the kinetic energy near the porous structure as its permeability increases, while maintaining a constant buoyant force. Remarkably, under particular scenarios and setups involving the permeable structure, the customary flow patterns undergo changes, providing valuable insights for engineers aiming to optimize the layout of clustered bodies for improved heat dissipation and storage effectiveness. This knowledge holds immense importance for professionals in the field.

Significant studies have illuminated the occurrence of heat and solute-driven motion in spaces filled with porous materials or passages, notably when a magnetic field comes into play. Furthermore, situations where heat and solute-driven scenarios emerge near grouped objects or permeable structures can be noticed in diverse practical uses. These include creating tiny electronic gadgets, controlling heat in nuclear reactors, making crystals, fuel cells, and cooling electronic devices. The effective use of permeable structures as capable devices for storing thermal energy is widely recognized due to their increased ability to conduct heat and their pore arrangement. To enhance energy retention, a commonly employed strategy entails incorporating the heat-storing apparatus as a solute origin. For instance, a clear case involves utilizing a reservoir of solute subjected to solar warming, demonstrating itself as a capable reservoir for storing energy, owing to adept confinement of heat-driven circulation [29]. Hence, the interaction between temperature and concentration disparities significantly affects the pace of circulation, presenting an avenue for meticulous manipulation of energy storage kinetics. On the flip side, the magnetic field has the ability

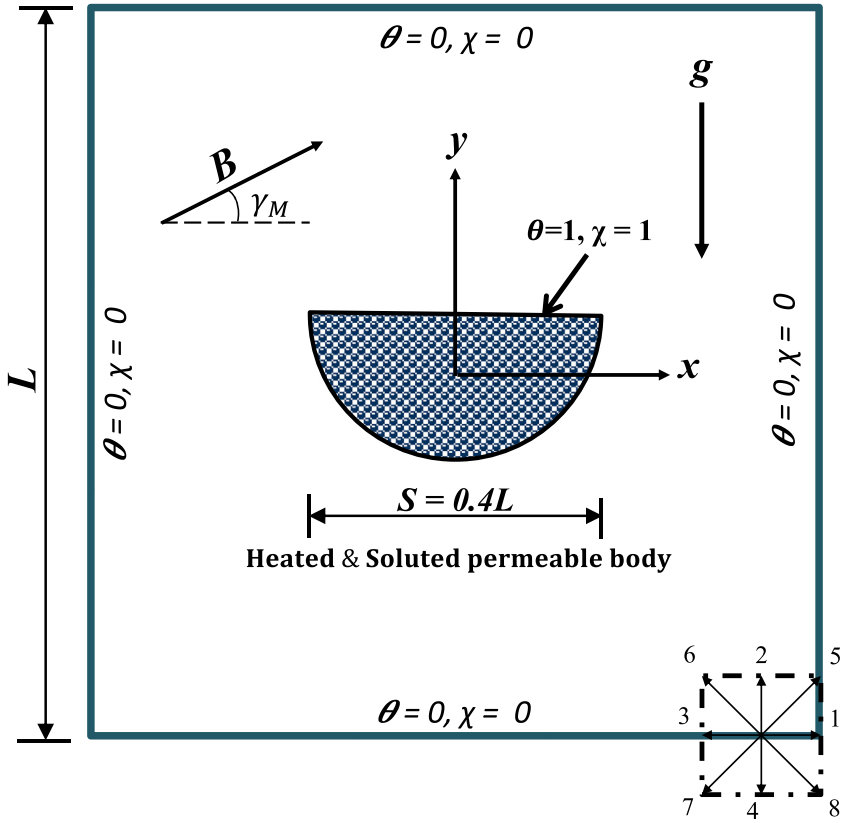


Fig. 1. Computational region for simulating the double-diffusive buoyancy-driven convection pattern around a permeable semi-circular body, subject to both heating and concentration, and further influenced by a magnetic field at an angle γ_M. The $0.4L$ times diameter of semi-circle's flat edge is facing upwards.

to influence the convection rate, satisfying a pivotal demand in contemporary utilities such as the creation of sophisticated electronic gadgets and energy storage setups. Furthermore, the amalgamation of planar and curved boundaries takes shape in the configuration of semi-circular cylinders, assuming distinct roles in the various applications examined earlier. When combined with permeability, the resultant shift in flow configurations, heat transmission, and mass transport consequences becomes exceptionally interesting. Through a more profound exploration of the complexities tied to heat and concentration movement from semi-circular structures made of porous materials, an opportunity emerges to tap into untapped potential for refining heat storage, dispersing thermal energy, and controlling concentration changes in the context of grouped structures configured in semi-circular forms. Despite its importance in various practical applications, as previously explained, the double-diffusion traits of these penetrable structures have yet to be fully investigated. The novelty of the work relies on the combined effect of semi-circular body with permeable traits under the influence of inclined magnetic field. Consequently, the present examination strives to unveil the elaborate dynamics of heat and mass transfer demonstrated by a permeable semi-circular structure in the presence of a magnetic field. Considering all these factors comprehensively, the authors have undertaken the current computational evaluation. The primary objective of this study is to assess the impact of permeability, the strength of the buoyancy ratio, the magnitude and direction of the magnetic force on the characteristics of double-diffusive natural convection encompassing the permeable semi-circular domain.

2. Mathematical framework

2.1. Problem delineation

The diagrammatic representation of the physical model utilized in this investigation is portrayed in Fig. 1. The model comprises a square enclosure with one side's length marked as L. Inside this clear fluid region enclosure, there exists a permeable semi-circular cylinder oriented with its flat edge facing upwards. The cylinder's diameter is denoted as S, where S is equal to 0.4 times the length of the enclosure ($S = 0.4L$). On the permeable section of the cylinder, constant high temperature T_h (where $T_h > T_c$) and high concentration C_h (where $C_h > C_c$) conditions are maintained. In contrast, the enclosure walls experience a constant low temperature T_c and low concentration C_c. Gravity acts in a downward direction, while a uniform magnetic force B (calculated as $\sqrt{B_x^2 + B_y^2}$) is applied at an angle γ_M, as depicted in Fig. 1. To facilitate numerical simulations, the following assumptions are adopted to ensure the model's practicality:

1. The fluid flow is considered to be two-dimensional and incompressible. Within the model, the permeable cylinder serves as an isotropic and homogeneous porous medium, displaying constant porous characteristics like permeability and porosity.
2. The driving force behind the fluid flow originates solely from the density difference resulting from the thermal and concentration gradients between the permeable semi-circular cylinder and the enclosure walls.
3. The imposed magnetic field exhibits significant dominance, leading to the neglect of any magnetic field effects caused by the motion of electrically conducting fluid. For simplicity, the thermal conductivity ratio and viscosity ratio of the porous medium are assumed to be the same as those of the fluid. Additionally, the permeable zone is subject to a local thermal equilibrium condition.

2.2. Conservation equations

The dimensionless macroscopic governing equations for the current investigation can be articulated in the following manner:

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0. \tag{1}$$

$$\frac{1}{\epsilon}\frac{\partial u}{\partial \tau_t}+\frac{1}{\epsilon^2}\left(u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}\right)=-\frac{\partial p}{\partial x}+Pr\frac{1}{\epsilon}\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right)-C_1\frac{Pr}{Da}u - C_2\frac{1.75}{\sqrt{150}}\frac{1}{\sqrt{Da}}\times\frac{\sqrt{u^2+v^2}}{\epsilon^{3/2}}u + \rho\frac{Ha^2 Pr}{\epsilon}(v\sin(\gamma_M)\cos(\gamma_M))-(u\sin^2(\gamma_M)). \tag{2}$$

$$\frac{1}{\epsilon}\frac{\partial v}{\partial \tau_t}+\frac{1}{\epsilon^2}\left(u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}\right) = -\frac{\partial p}{\partial y}+Pr\frac{1}{\epsilon}\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}\right)-C_1\frac{Pr}{Da}v-C_2\frac{1.75}{\sqrt{150}}\frac{1}{\sqrt{Da}}\times\frac{\sqrt{u^2+v^2}}{\epsilon^{3/2}}v + \rho\frac{Ha^2 Pr}{\epsilon}(u\sin(\gamma_M)\cos(\gamma_M))-(v\cos^2(\gamma_M))+Pr\ Ra(\theta+BR\ \chi). \tag{3}$$

$$\frac{1}{\epsilon}\left(\frac{\partial \theta}{\partial \tau_t}+u\frac{\partial \theta}{\partial x}+v\frac{\partial \theta}{\partial y}\right)=\frac{1}{Pr}\left(\frac{\partial^2\theta}{\partial x^2}+\frac{\partial^2\theta}{\partial y^2}\right). \tag{4}$$

$$\frac{1}{\epsilon}\left(\frac{\partial \chi}{\partial \tau_t}+u\frac{\partial \chi}{\partial x}+v\frac{\partial \chi}{\partial y}\right)=\frac{1}{Le}\left(\frac{\partial^2\chi}{\partial x^2}+\frac{\partial^2\chi}{\partial y^2}\right). \tag{5}$$

In the aforementioned equations, the values of C_1 and C_2 are set to 1 within the porous zone, while they are assigned a value of 0 in the clear fluid zone. These assignments are made in the context of dimensionless variables, which include the Prandtl number denoted as Pr (calculated as $(\mu C_p)/k$), the Lewis number denoted as Le (calculated as α/D), the Darcy number denoted as Da, and the Hartmann number denoted as Ha. It is important to note that these dimensionless equations have been derived using the following relationships: $x=\frac{x^*}{L}$, $y=\frac{y^*}{L}$, $p=\frac{p^*L^2}{\rho\alpha^2}$, $u=\frac{u^*L}{\alpha}$, $v=\frac{v^*L}{\alpha}$, $\tau_t=\frac{t^*\alpha}{L^2}$ $\theta=\frac{T-T_\infty}{T_w-T_\infty}$, $\chi=\frac{C-C_\infty}{C_w-C_\infty}$.

It is worth noting that the superscript * is employed to signify the dimensional form of the variable.

2.3. Lattice Boltzmann method (LBM)

The current study utilizes the lattice Boltzmann (LB) technique to model the combined effects of double-diffusion, magnetic fields, and porous media characteristics. The remarkable simplicity of this kinetic approach has inspired researchers to employ it in various fields. Notably, LBM has demonstrated its effectiveness in handling flow through porous media, complex geometries, and multi-phase flows. This technique involves modelling the flow by analysing the arrangement of grouped particles. The sequential steps in LBM include collision, streaming, application of boundary conditions, and evaluation of macroscopic characteristics. Subsequently, the resulting macroscopic properties are translated into distinct distribution functions, facilitating the acquisition of thermal and concentration fields.

2.3.1. LB governing equations for flow field

The collision equation within the flow field takes into account the direct influence of force terms linked to buoyant force, magnetic field, and porous medium. It can be represented in the following manner [20,63]:

$$f_i(x+e_i\Delta t,t+\Delta t)-f_i(x,t)=-\frac{1}{\tau}\left[f_i(x,t)-f_i^{eq}(x,t)\right]+\Delta t F_i+\Delta t F_b+\Delta t F_M. \tag{6}$$

In Eq. (6), the symbols f_i, f_i^{eq}, and e_i represent the particle distribution function, equilibrium distribution function, and the particle velocity vector for the ith link, respectively. The analysis utilizes the $D2Q9$ lattice arrangement, as depicted in Fig. 1. The velocity of each particle in this arrangement ($\Delta x_i/\Delta t$) is presented below.

$$e_i=\begin{cases}(0,0), & i=0\\ (cos[(i-1)\pi/2],sin[(i-1)\pi/2])e, & i=1-4\\ (cos[(2i-9)\pi/4],sin[(2i-9)\pi/4])\sqrt{2}e, & i=5-8.\end{cases} \tag{7}$$

In this particular study, the BGK-collision operator with a single relaxation time (SRT) is put into use to assist in the relaxation of particles towards their equilibrium state. To achieve this, a solitary relaxation factor represented by τ is utilized. The connection between this relaxation factor and the kinematic viscosity, denoted as ν, is established through the Chapman-Enskog relationship, and it can be expressed in the following manner:

$$\nu=\left(\tau-\frac{1}{2}\right)c_s^2\Delta t, \tag{8}$$

Within the context of Eq. (8), the variable c_s^2 signifies the speed of sound in lattice, which equals $e/\sqrt{3}$. In order to incorporate the viscous and inertial effects of the porous medium, a force term denoted as F_i is introduced alongside the BGK collision operator. Thus, taking into account the porosity (ϵ) of the porous medium, the expression for F_i is as follows:

$$F_i=w_i\rho\left(1-\frac{1}{2\tau}\right)\left[3(e_i.F)+\frac{9}{\epsilon}(e_i.U)(e_i.F)-\frac{3}{\epsilon}(U.F)\right]. \tag{9}$$

The force term F in this context represents the Darcy–Forchheimer force, and its mathematical expression can be written as follows:

$$F=-\frac{\epsilon\nu}{K}U-\frac{\epsilon c_F}{\sqrt{K}}|U|U+\epsilon G. \tag{10}$$

In the above expression, the variable K denotes the permeability of the porous medium. Additionally, the non-dimensional Forchheimer term, denoted by c_F (introduced by Fu et al. [64]) is equal to $1.75/(\sqrt{150\times\epsilon^3})$. In Eq. (6), the equilibrium distribution function f_i^{eq} must consider the porosity within the porous region while maintaining a value of unity in the clear fluid region. The expression for f_i^{eq} is as follows:

$$f_i^{eq}=w_i\rho\left[1+3(e_i.U)+\frac{9}{2\epsilon}(e_i.U)^2-\frac{3}{2\epsilon}(U.U)\right]. \tag{11}$$

The particles' weighting factors, denoted as w_i, take specific values: $w_0=4/9$, $w_i=1/9$ for i within the range of 1 to 4, and $w_i=1/36$ for i within the range of 5 to 8. During each time step, the computed macroscopic density and velocity values are transferred to equations Eq. (9)–(11). These equations are derived from the distribution functions f_i. The macroscopic properties are determined using the expressions below:

$$\rho=\sum_{i=0}^{8}f_i. \tag{12}$$

The auxiliary velocity, denoted as V, is calculated as

$$\rho V=\sum_{i=1}^{8}e_i f_i+\frac{\Delta t}{2}\rho F. \tag{13}$$

The velocity component in the porous medium is derived from the auxiliary velocity and can be represented as:

$$U=\frac{V}{c_0+\sqrt{c_0^2+c_1|V|}}. \tag{14}$$

The terms c_0 and c_1 in Eq. (14) can be obtained as follows:

$$c_0 = \frac{1}{2}\left(1 + \epsilon\frac{\Delta t}{2}\frac{\nu}{K}\right); \qquad c_1 = \epsilon\frac{\Delta t}{2}\frac{c_F}{\sqrt{K}}. \tag{15}$$

In scenarios involving double diffusion, where the flow field is influenced by gradients in temperature and concentration, the dominant effects primarily depend on the force of gravity (g). As a result, the buoyant force term (F_b) can be mathematically expressed as:

$$F_b = 3w_i\ g\ e_{iy}(\beta_T\theta\ + \beta_C\chi). \tag{16}$$

Here, the symbols g, β_C, and β_T represent the gravitational force due to acceleration, concentration coefficient, and thermal expansion coefficient, respectively. The term e_{iy} denotes the velocity vectors in the y-direction. Additionally, the buoyancy ratio is defined as $BR = (\beta_C\Delta C)/(\beta_T\Delta T)$. Moreover, the force term that considers the effects of the magnetic field on the flow field, denoted as $F_M = F_{MX} + F_{MY}$ can be expressed as

$$F_{Mx} = 3w_i\rho\left[A\left((v\ \sin(\gamma_M)\ \cos(\gamma_M)) - (u\ \sin^2(\gamma_M))\right)\right], \tag{17}$$

$$F_{My} = 3w_i\rho\left[A\left((u\ \sin(\gamma_M)\ \cos(\gamma_M)) - (v\ \cos^2(\gamma_M))\right)\right]. \tag{18}$$

Within the provided equation, the parameter A is defined as $Ha^2\mu/L^2$, where Ha represents the Hartmann number defined as $LB_0\sqrt{\sigma}\mu$, and γ_M denotes the angle of the magnetic field concerning the x-direction.

2.3.2. LB governing equations for scalar field

To investigate the scalar field (either temperature or concentration), an additional distribution function denoted as Π_i is used. This distribution function is subsequently replaced by g for thermal field and C for concentration field. The evolution equation governing the scalar field can be expressed as:

$$\Pi_i(x + e_i\Delta t, t + \Delta t) - \Pi_i(x,t) = -\frac{1}{Y}\left(\Pi_i(x,t) - \Pi_i^{eq}(x,t)\right). \tag{19}$$

In Eq. (19), the non-dimensional relaxation time for the scalar field is represented as Y. However, for the temperature and concentration fields, the respective values τ' and τ'' are employed. Additionally, these values are linked to the thermal diffusion coefficient ($\alpha = (\tau' - 1/2)c_s^2\Delta t$) and the concentration diffusion coefficient ($D = (\tau'' - 1/2)c_s^2\Delta t$).

$$\Pi_i^{eq} = w_i\eta[1 + 3(e_i.U)]. \tag{20}$$

Here, η represents the non-dimensional scalar field, which can be calculated from the distribution function Π_i using the following expression:

$$\eta = \sum_{i=0}^{8}\Pi_i. \tag{21}$$

In Eq. (21), the variable η is substituted by θ to represent the temperature field, and by χ to represent the concentration field, respectively.

2.4. LB boundary specifications

At the enclosure walls, the velocity field adheres to a no-slip boundary condition, specifically $u = 0$ and $v = 0$. For the temperature field, a condition of low temperature ($\theta = 0$) is enforced, while the concentration field is subjected to a low concentration condition ($\chi = 0$). Inside the porous cylinder, a force term denoted as F_i is employed to mimic the porous characteristics in the velocity field. Moreover, the temperature field experiences a high temperature condition ($\theta = 1$), and correspondingly, the concentration field is set to a high concentration level ($\chi = 1$). In the context of LBM, the boundary conditions are established based on distribution functions. At the walls of the enclosure, a bounce-back boundary condition is applied. This condition sets the distribution functions moving from the wall into the fluid zone equal to their corresponding opposite distribution functions. For example, on the top wall of the enclosure, the unknown distribution functions f_4, f_7, and f_8 are equated to f_2, f_5, and f_6, respectively. Similar boundary conditions can be defined for other walls in a similar manner. Additionally, high temperature and high concentration boundary conditions are consistently imposed within the porous cylinder, specified as follows:

$$g_i + \tilde{g}_i = \theta \times (w_i + \tilde{w}_i), \tag{22}$$

$$C_i + \tilde{C}_i = \chi \times (w_i + \tilde{w}_i). \tag{23}$$

Inside the porous cylinder, the variables θ, χ, and $\tilde{w}_i$ in the mentioned equations are assigned a value of 1. Additionally, $\tilde{g}$ and $\tilde{C}$ represent the opposite distribution function of temperature and concentration, respectively, for the lattice link i. On the walls of the enclosure, the boundary conditions for low temperature and low concentration are imposed by setting θ and χ to 0 in the mentioned equations.

2.5. Dimensionless parameters

The determination of non-dimensional parameters in this study is based on the ratio of lattice units (N) to the characteristic length (L). The non-dimensional parameters used in the current research are elaborated as follows:

The expression representing the Rayleigh number (Ra) is formulated as follows:

$$Ra = \frac{g\beta\Delta T L^3}{\alpha\nu} = \frac{g\beta\Delta T N^3}{\alpha\nu}. \tag{24}$$

Porosity (ϵ) and permeability (K) are fundamental indicators of porous medium properties. The non-dimensional representation of permeability is referred to as the Darcy number (Da). In the current investigation, the width of the porous medium corresponds to the diameter (S) of the semi-circular porous body. Hence, the expression for Da can be given as follows:

$$Da = \frac{K}{S^2}. \tag{25}$$

The Carman–Kozeny relationship [65] establishes a correlation between Da and ϵ for a densely packed sphere bed. This correlation can be expressed by the following equation:

$$Da = \frac{1}{180}\frac{\epsilon^3 d_{pp}^2}{W^2(1-\epsilon)^2}, \tag{26}$$

In this context, d_{pp} denotes the diameter of the spherical particles within the porous medium. It is noteworthy that by varying d_{pp}, different combinations of Da and ϵ values can be achieved. For the purpose of this study, a value of 1 cm is adopted for d_{pp}, corresponding to a width of 1 m for the porous medium.

3. Computational method and verification

This section outlines the computational method, grid sensitivity test, and the process of confirming the accuracy of the current code's results through comparison with published data. The numerical procedure provides detailed explanations regarding the incorporation of the porous zone activation into the LB governing equation, the determination of LB properties, and the establishment of convergence criteria.

3.1. Computational technique

To conduct the present numerical investigation, a lattice Boltzmann code is developed using the FORTRAN programming language. The code performs a series of sequential operations in the lattice Boltzmann method (LBM), including collision, streaming, handling boundary conditions, and computing macroscopic properties. Before entering the main loop, the LB viscosity (ν), thermal diffusivity (α), and mass diffusivity (D) are calculated based on the values of Ra, Pr, and Le, respectively. Moreover, non-dimensional relaxation factors for flow (τ), temperature (τ'), and concentration (τ'') are derived using ν, α, and D, respectively. The collision equation for the flow field, incorporating

Table 1
Mesh sensitivity study on mean Nusselt number (Nu_M), and mean Sherwood number (Sh_M) of the enclosure and overall entropy values (S_T) in the enclosure for $BR = 5$, $Da = 10^{-3}$, & $Ha = 0$.

Mesh size	Nu_M	% deviation	Sh_M	% deviation
150 × 150	3.97944	1.7671	10.38140	2.8014
250 × 250	4.02873	0.5221	10.58602	0.8143
300 × 300[a]	4.05266	0.0715	10.68622	0.1311
400 × 400	4.04976	–	10.67223	–

[a] Mesh size employed in the analysis.

magnetic intensity, porous characteristics, and buoyant force, is applied throughout the computational domain. However, to account for the effects of porous characteristics in the clear fluid region, the force term F_i in Eq. (6) is multiplied by a value of zero. Subsequently, the macroscopic properties of the flow are evaluated using equations Eq. (12) and (13), followed by the computation of the scalar collision equation (Eq. (19)) at each node. This is then followed by the calculation of macroscopic temperature and concentration values. In this particular investigation, a value of 0.1 is assigned to $\sqrt{g(\beta_T \Delta\theta + \beta_C \Delta\chi)N}$ to mitigate compressibility effects. It is essential to highlight that this factor maintains a direct correlation with the characteristic velocity. When the value of $\sqrt{g(\beta_T \Delta\theta + \beta_C \Delta\chi)N}$ dips below 0.1, it triggers a proportional uptick in computational time. Conversely, when this term reaches exceedingly low values, it greatly bolsters the stability of the simulation. Equation Eq. (11) demonstrates that the current numerical scheme exhibits second-order accuracy. The iterations continue until the specified convergence criteria are met.

$$\frac{\sqrt{\sum_{i,j}\left[U_{i,j}^{(k)} - U_{i,j}^{(k-100)}\right]^2}}{\sqrt{\sum_{i,j}\left[U_{i,j}^{(k)}\right]^2}} \leq 10^{-6}, \quad \frac{\sqrt{\sum_{i,j}\left[\theta_{i,j}^{(k)} - \theta_{i,j}^{(k-100)}\right]^2}}{\sqrt{\sum_{i,j}\left[\theta_{i,j}^{(k)}\right]^2}} \leq 10^{-6},$$
$$\text{and} \quad \frac{\sqrt{\sum_{i,j}\left[\chi_{i,j}^{(k)} - \chi_{i,j}^{(k-100)}\right]^2}}{\sqrt{\sum_{i,j}\left[\chi_{i,j}^{(k)}\right]^2}} \leq 10^{-6}. \tag{27}$$

Here, $U_{i,j}$, $\theta_{i,j}$, $\chi_{i,j}$, and k are the fluid velocity, temperature, concentration, and iteration level, respectively.

3.2. Mesh refinement analysis

To assess the robustness and precision of our computational model, the mesh independence tests, scrutinizing the deviations of crucial parameters within the enclosure are conducted. Key variables under scrutiny include the mean Nusselt number (Nu_M), and mean Sherwood number (Sh_M) of the enclosure walls. In the quest for reliable results, these tests are carefully executed under specific conditions: $BR = 5$, $Da = 10^{-3}$, and $Ha = 0$, which correspond to scenarios with significant temperature and/or concentration gradients. The stakes are high, particularly when dealing with elevated values of Da and BR, making grid reliability a paramount concern. Considering the magnetic field's inhibiting effect on heat and concentration transfer, $Ha = 0$ is opted. Table 1 indicates that among varying lattice numbers, the illustrious 300 × 300 grid is consistently delivering precise results while ensuring computational efficiency. In LB technique, lattice spacing (Δx) and time step (Δt) are equal, and it's value kept unity in this study.

3.3. Code verification

The effectiveness of the current LBM programming script is evaluated by conducting a comparative analysis involving Nu_M, and Sh_M, which are then compared against published data. Initially, we compare the values of Nu_M and Sh_M with those presented by Kefayati and Tang [47] in a scenario involving double-diffusive buoyancy-driven convection within a heated container that houses an inner cold circular cylinder. This comparison spans various combinations of BR and Ra values while maintaining constant values of $Pr = 0.1$ and $Le = 2.5$ (Refer Table 2). Subsequently, in Fig. 2(a), we illustrate the comparison of Nu_M and Sh_M concerning MHD double-diffusive free-convection unfolding within a rectangular enclosure. This presentation compares the results against those documented by Chamkha and Al-Naser [17] for a range of Ha values. The parameters are held constant at $Gr = 10^5$, $Le = 2$, $BR = -0.8$, aspect ratio of 2, and $Pr = 1$. In order to confirm the precise representation of double diffusion phenomena in permeable materials by the LBM programming script, we assess the agreement of Sh_M and Nu_M values with those reported by Mohamad and Bennacer [29]. This evaluation focuses on the context of thermo-solutal free convection taking place within a container saturated with a porous medium. This comparison is detailed in Fig. 2(b) and includes various BR values under the conditions of $Le = Pr = 10$, $Da = 10^{-3}$, and $Ra = 10^5$. The analysis indicates that the current program shows a percentage deviation of fewer than 2% when contrasted with the prevailing published data. On the whole, the outcomes suggest that the present LBM code performs effectively, aligning closely with the literature in relation to the assessed parameters.

4. Results and discussion

In this study, computer simulations serve as the gateway to explore the complexities of two-dimensional double-diffusive natural convection. These simulations delve deeply into the interaction between a heated and concentrated permeable semi-circular body and a square enclosure. Moreover, the magnetic field's influence is thoughtfully incorporated to understand its impact on flow, heat, and concentration phenomena. The following array of parameters has been considered in this pursuit:

- Darcy number, $Da = 10^{-5}$, 10^{-4}, and 10^{-3}.
- Porosity, $\epsilon = 0.824$, 0.935, and 0.977.
- Buoyancy ratio, $BR = -5$, -2, -1, 0, 1, 2 and 5.
- Hartmann number, $Ha = 0$, 25, and 50.
- Magnetic angle, $\gamma_M = 0°$, 45°, and 90°.

The present investigation encompasses Da values spanning from 10^{-5} to 10^{-3}, encompassing the high to low viscous resistance within a porous medium. This range is considered to accentuate the distinctiveness of porous characteristics and their consequential impact on double-diffusion phenomena. In accordance with the relationship denoted as Eq. (26), the assessment of ϵ values are done. Furthermore, the introduction of a magnetic field exerts a profound influence on the thermo-solutal patterns, particularly as Ha escalates from 0 to 50, after which it becomes trivial for $Ra = 10^5$. Additionally, the investigation scrutinizes the variation of γ_M as it transitions from a horizontal to a vertical orientation, thereby encompassing the entire spectrum of angular effects. Conversely, the buoyancy force attributed to concentration is magnified to fivefold that of the temperature buoyancy force, under both opposing and aiding conditions. Within this ambit of buoyancy ratio (BR), the cumulative effect of these buoyancy forces emerges as a significant factor, particularly when $Le = 5$. As the buoyancy ratio (BR) continues to increase, it becomes evident that the dominant influence is exerted by the concentration buoyancy force alone. By considering all these factors, the above ranges of parameters are considered in the analysis. Throughout this comprehensive study, the values of Ra, Pr, and Le stand resolute at 10^5, 7.0, and 5, respectively.

4.1. Hydro-thermal and concentration traits

In this section, an in-depth exploration of various factors on the hydro-thermal and concentration characteristics takes center stage. Employing a blend of quantitative and qualitative analysis, the profound

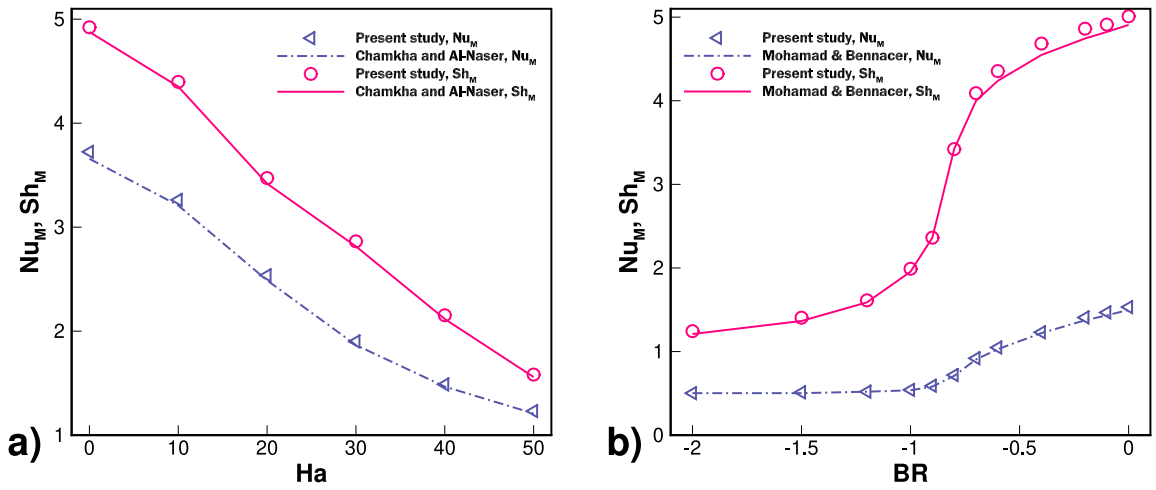


Fig. 2. Analogy of Nu_M&Sh_M for the scenario of thermo-solutal buoyancy-driven heat transfer (a) on the influence of magnetic effect within a rectangular enclosure from the present code with Chamkha and Al-Naser [17] for various values of Ha at $Gr = 10^5$, $Le = 2$, $BR = -0.8$, aspect ratio= 2, and $Pr = 1$, and (b) on the influence of porous filled enclosure from the present code with Mohamad and Bennacer [29] for various values of BR at $Da = 10^{-3}$, $Pr = 10$, $Ra = 10^5$, and $Le = 10$.

Table 2
Comparison of Nu_M&Sh_M for the scenario of thermo-solutal buoyancy-driven convection in a heated enclosure with an inner cold circular body with [47] for various values of BR&Ra at $Pr = 0.1$&$Le = 2.5$.

Ra	BR	Nu_M			Sh_M		
		Present study	[47]	% deviation	Present study	[47]	% deviation
10^4	−1	6.3104	6.2455	1.040	5.7772	5.7091	1.193
	0.1	6.3671	6.3267	0.638	6.1843	6.1284	0.912
	1	6.5443	6.5005	0.674	7.0124	6.9565	0.804
10^5	−1	5.3047	5.2679	0.699	2.1534	2.1248	1.344
	0.1	6.8067	6.7434	0.938	5.2671	5.2232	0.840
	1	8.5221	8.4213	1.197	9.1553	9.0629	1.020

effects of the Darcy number (Da), Buoyancy ratio (BR), Hartmann number (Ha), and magnetic field angle (γ_M) are investigated. Each parameter is subjected to scrutiny, offering valuable insights into their impact on the dynamic behaviour of heat and concentration transfer. Notably, within the streamline pattern, the solid and dotted lines symbolize counter-clockwise and clockwise fluid flow directions, respectively. To ensure utmost clarity, the isotherm and iso-concentration patterns are shaded with contours ranging from 1 to 0.9.

4.1.1. Significance of Da

Figs. 3(a)–(c) expose the influence of dimensionless permeability (Da) on the flow patterns, temperature distribution, and concentration profiles for the condition of $BR = 2$, $\gamma_M = 90°$, and $Ha = 25$. When Da reaches the value of 10^{-5}, the porous body demonstrates a notable resistance to fluid flow, which is clearly observable from the streamline pattern. At this particular Da value, the porous body experiences significant viscous resistance, effectively impeding fluid motion within the porous region. Consequently, the fluid loses a considerable amount of its kinetic energy in the permeable region at this Da. An important observation is that the fluid path within the porous zone converges from penetration to the exit end. This converging flow pattern is influenced by the curved fore-body, which creates the least porous zone as the fluid enters. On the other hand, when the fore-body has a flat edge, a divergent flow pattern is observed. The isothermal and iso-concentration lines become narrower towards the top portion of the enclosure, closely following the curved edge of the semi-circular cylinder. Due to Le exceeding unity, the concentration gradient around the cylinder becomes more pronounced compared to the thermal gradient. In simpler terms, increasing the value of Le enhances the thickness of the thermal boundary layer compared to the concentration boundary layer, resulting in improved concentration transfer. As a result, adjusting the value of Le in energy storage devices can enhance their thermal storage capacity. As Da increases to 10^{-4}, a small amount of fluid infiltrates into the cylinder, resulting in a flow pattern similar to that observed at lower Da values. However, due to the limited flow through the porous body, the isotherms and iso-concentration lines become slightly elongated. From Figs. 3(a)&(b), it becomes apparent that the stream function value, which previously exhibited penetration within the extreme portion of the porous zone at $Da = 10^{-5}$, shifts towards the center portion of the semi-circular body as Da transitions to 10^{-4}. When Da reaches 10^{-3}, the viscous resistance provided by the porous medium undergoes a remarkable decrease, allowing a substantial volume of flow to pass through the cylinder. As the heated and concentrated fluid emerges from the flat edge of the permeable body, the thermal and concentration gradients in that area gradually diminish. At the same time, the thermal and concentration boundary layers become thinner around the curved portion of the porous body due to the increased inflow of colder fluid. As a result, enhancing the permeability of a thermal device leads to improved heat delivery. This occurs because the reduced viscous resistance and thinner boundary layers allow for more efficient heat transfer and concentration dispersion within the system. Moreover, as Da increases, the flow intensity around the porous cylinder intensifies. This intensification is driven by the elevated thermal and concentration gradients, which locally amplify the buoyant force, thus fuelling the enhancement in the flow field. In summary, permeability plays a crucial role in boosting heat and concentration transfer within the system. Additionally, it augments the kinetic energy of the fluid in the vicinity of the permeable zone, resulting in a more vigorous and efficient fluid motion.

4.1.2. Significance of BR

Figs. 4(a)–(c) exemplify the influence of the buoyancy ratio (BR) on the flow behaviour, temperature dispersion, and concentration distribution under specific conditions: $Da = 10^{-3}$, $\gamma_M = 90°$, and $Ha =$

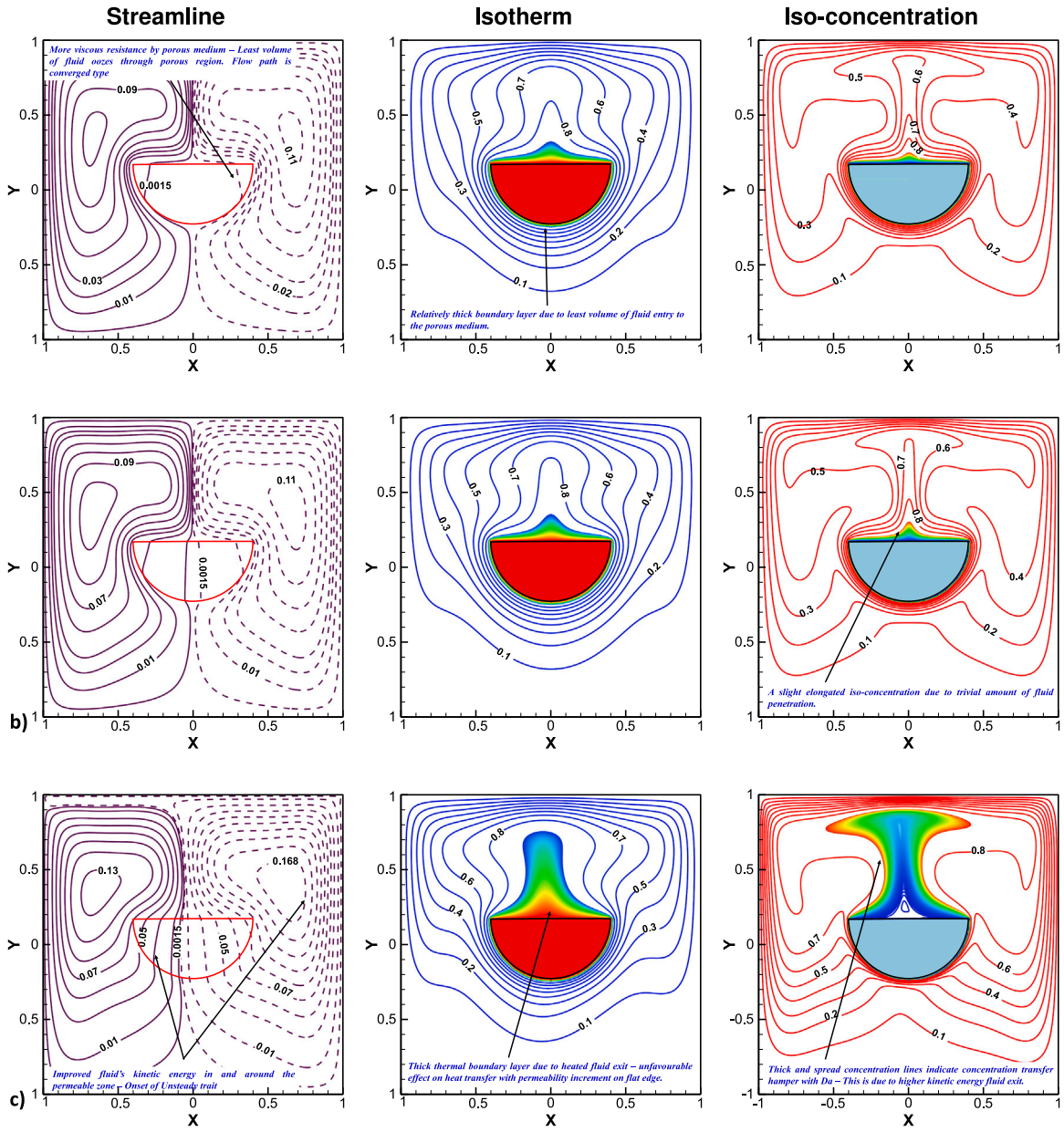


Fig. 3. Significance of the Darcy number (Da) on the configuration of streamlines, isotherms and iso-concentration lines under the conditions of $Ha = 25$, $BR = 2$, and $\gamma_M = 90°$, displayed for (a) $Da = 10^{-5}$, (b) $Da = 10^{-4}$, and (c) $Da = 10^{-3}$.

25. When BR is set to -5, the fluid encounters a downward thrust due to the buoyant force, causing a discernible deviation of negative streamlines towards the left boundary of the enclosure. When BR takes a negative value, it induces a reversal in the gravitational force, causing the low-density fluid to move downward, influenced by both heat and concentration effects. As a consequence of this shift, the fluid tends to migrate towards the flat side of the semi-circular body, leading to the breaking of the hydrodynamic boundary layer at the corners. Additionally, the buoyant force arising from thermal gradients promotes an upward motion of the fluid. The combination of these phenomena gives rise to distinct unsteady characteristics, clearly evident from the asymmetric contour profile shown in Fig. 4(a). The dominance of unsteadiness is particularly evident in the concentration contours rather than the isotherms, underscoring that the concentration transfer is more pronounced compared to the heat transfer. Moreover, the iso-concentration lines adhere more closely to the flat edge of the semi-circular cylinder in contrast to the isotherms. As BR increases from -5, the stream function value progressively weakens until it reaches -1. When $BR = -1$, the stream function values are observed to be at their minimum compared to all other BR values, leading to a reduction in heat and concentration transfer rates. This occurrence emerges due to the equilibrium reached between the thermal buoyant force and concentration buoyant force, with both exerting opposing effects. However, considering that the Lewis number (Le) is greater than 1, the concentration buoyant force prevails significantly over the thermal buoyant force. Consequently, the fluid tends to flow towards the bottom wall of the enclosure. At $BR = 0$, the fluid flow is exclusively influenced by the temperature gradient since there is no concentration buoyant force present. Consequently, the kinetic energy of the fluid within and around the porous zone is reduced. This, in turn, leads to a decrease in the thermal and concentration gradients around the semi-circular body. Additionally, the widening of temperature and concentration contours, resulting from the fluid exiting the flat edge with lower kinetic energy, indicates a decline in both heat and concentration transfers. For positive BR values, the concentration diffusivity enhances the overall buoyant force, leading to an escalation in the flow intensity as BR

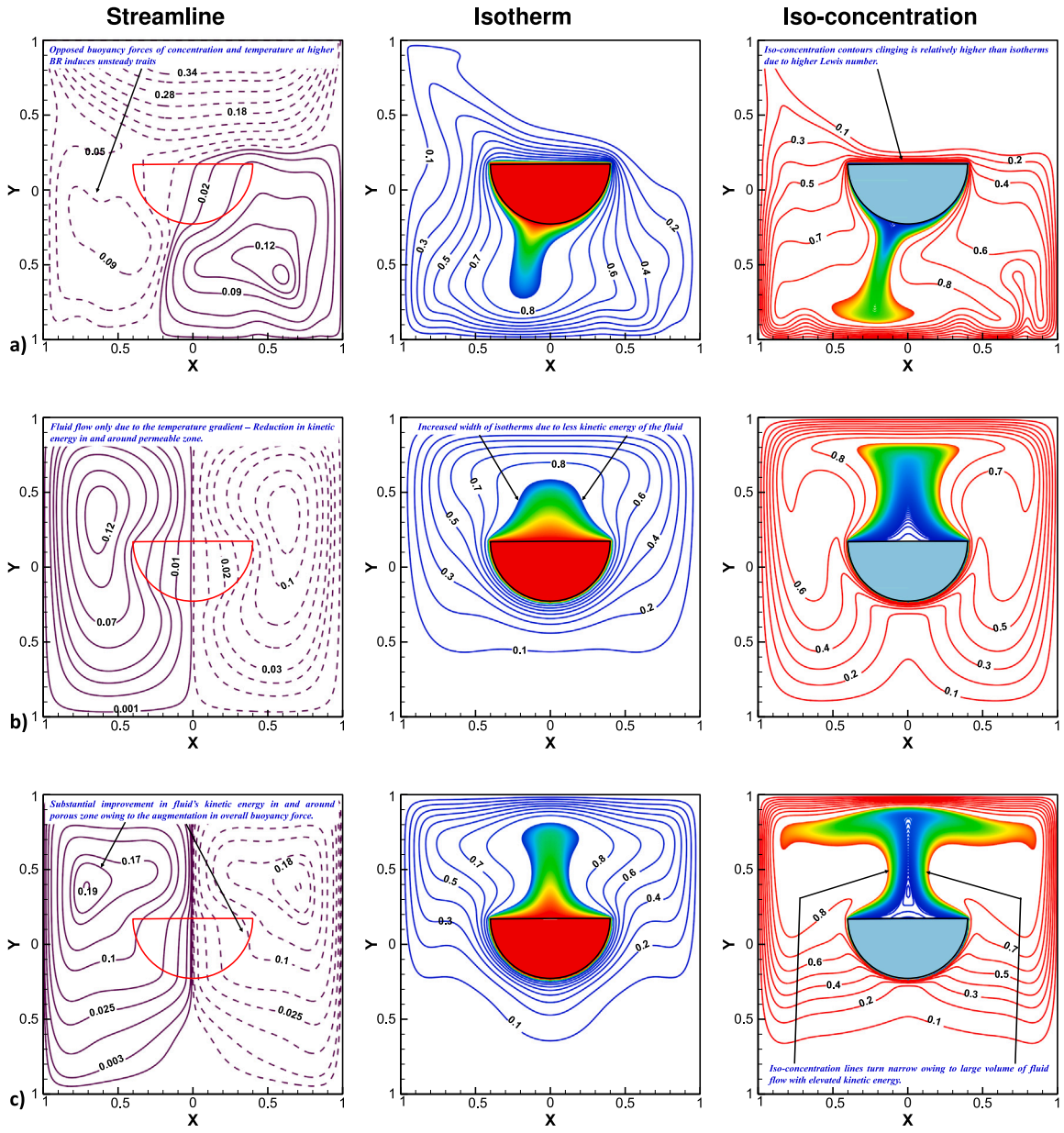


Fig. 4. Significance of the buoyancy ratio (BR) on the configuration of streamlines, isotherms and iso-concentration lines under the conditions of $Ha = 25$, $Da = 10^{-3}$, and $\gamma_M = 90°$, displayed for (a) $BR = -5$, (b) $BR = 0$, and (c) $BR = 5$.

increases. Consequently, a larger volume of fluid with higher kinetic energy is driven into the porous cylinder. This causes the thermal and concentration boundary layers to thicken in the middle region of the flat edge of the semi-circular cylinder, while simultaneously increasing the thermal and concentration gradients along the curved regions of the porous body. In conclusion, to enhance the heat and mass transfer characteristics, two approaches can be employed. First, increasing the temperature and concentration difference between the fluid and the surrounding environment would boost the transfer rates. Second, keeping the temperature difference constant and elevating the concentration difference would strengthen the total buoyant force, resulting in an improvement in thermo-solutal transfer.

4.1.3. Significance of Ha

The impact of the magnetic field on the streamline, isotherm, and iso-concentration patterns is demonstrated in Figs. 5(a)–(c) under the specified conditions of $BR = 2$, $Da = 10^{-3}$, and $\gamma_M = 45°$. In the absence of a magnetic field ($Ha = 0$), the stream function values are notably higher compared to other cases with non-zero Ha. It is important to note that the magnetic field mitigates the influence of BR on the enhancement of flow field intensity, particularly in the vicinity of the semi-circular cylinder. This effect becomes evident when comparing Figs. 4 and 5. The isotherms and iso-concentration lines display a slender elongation in the y-direction, signifying pronounced temperature and concentration gradients around the semi-circular cylinder, except for the topmost region. As the magnetic field strength intensifies, the flow field undergoes a reduction in momentum, causing the isotherm and iso-concentration lines to spread out in the streamwise direction. This phenomenon is clearly evident from the analysis of equations Eq. (17) and (18), which demonstrate that higher values of Ha lead to an increased magnitude of velocity in the opposite direction. As a result, the magnetic force (F_M) becomes negative, leading to a reduction in thermal and concentration gradients around the porous cylinder. It is worth noting that the contours appear slanted due to setting $\gamma_M = 45°$. Indeed, when Ha equals zero, the magnetic angle has no noticeable effect. Additionally, an increase in Ha exhibits

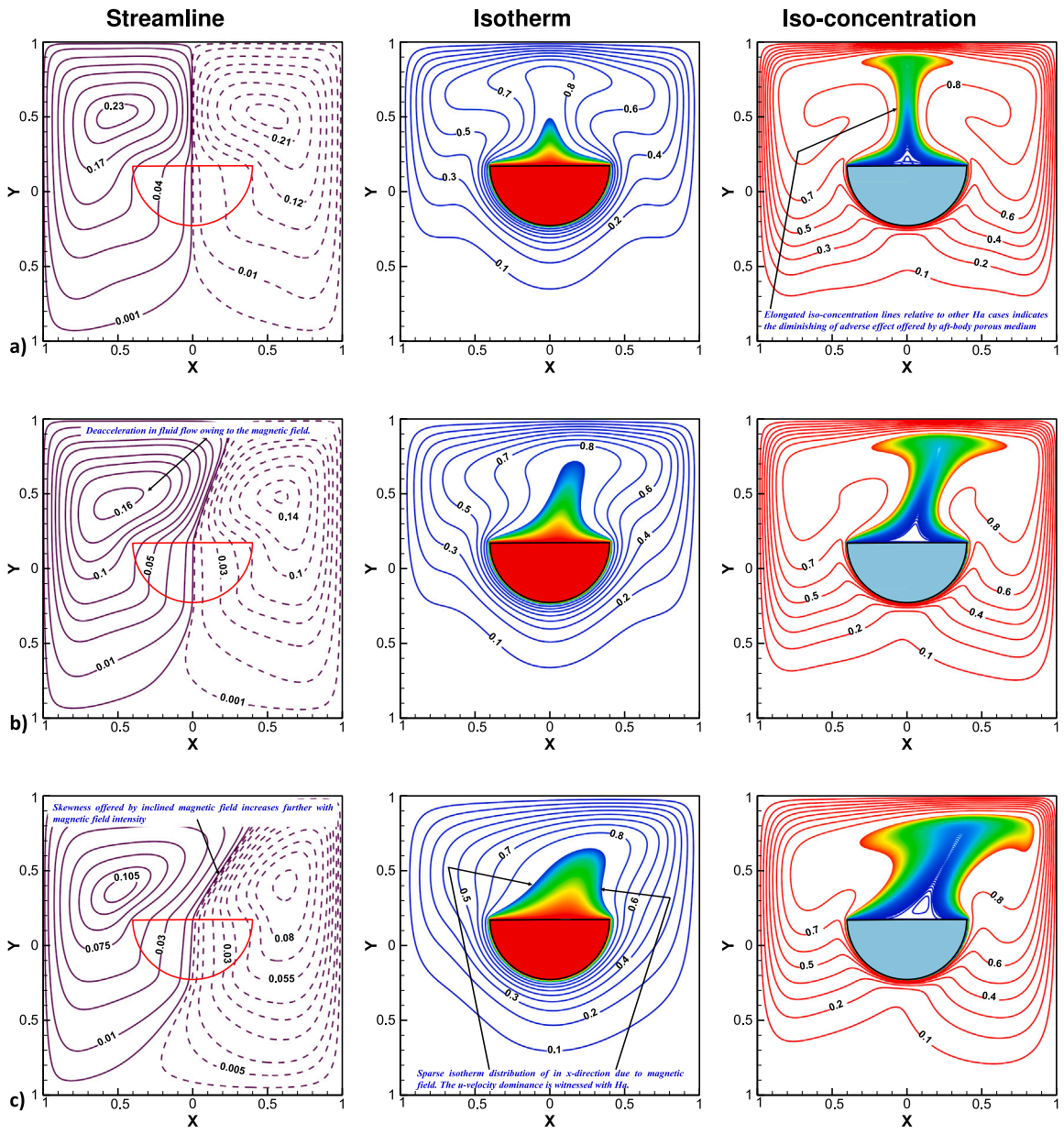


Fig. 5. Significance of the Hartmann number (Ha) on the configuration of streamlines, isotherms and iso-concentration lines under the conditions of $\gamma_M = 45°$, $BR = 2$, and $Da = 10^{-3}$, displayed for (a) $Ha = 0$, (b) $Ha = 25$, and (c) $Ha = 50$.

only a negligible impact on the flow field intensity within the semi-circular porous zone, despite inducing significant variations around it. One crucial point to emphasize is that the velocity values within the porous zone are lower compared to the region occupied by clear fluid. Consequently, the term representing the force exerted by the magnetic field undergoes minimal changes due to the subdued flow intensity within the porous zone.

4.1.4. Significance of γ_M

Figs. 6(a)–(c) provides a visual representation of the influence of γ_M on the patterns of streamlines, isotherms, and iso-concentrations to the specific conditions of $BR = 2$, $Da = 10^{-3}$, and $Ha = 25$. Notably, the contours exhibit a slanted appearance only when γ_M is set to 45 degrees, irrespective of the parameter values explored in this study. Moreover, this skewness becomes more pronounced with increasing Ha, causing notable changes in the thermal and concentration gradients around the permeable cylinder (see Fig. 5(c)). Additionally, elevating γ_M from 0 to 45 degrees leads to a slight improvement in the intensity of the flow field. However, a substantial intensification occurs when γ_M shifts to 90 degrees, triggering unsteady characteristics due to the increased kinetic energy. As a result, the heat and concentration transfer rates escalate with the increasing magnetic angle. When the magnetic field is oriented upwards, the x-directional velocity is suppressed, causing the streamlines to be predominantly influenced by the y-directional velocity (as seen in Eqs. (17) and (18)). Consequently, fluid flow movement becomes enriched in the y-direction, facilitating higher rates of heat and concentration transfer. It is intriguing to observe that the heightened flow field intensity gives rise to an unsteady flow pattern, which becomes more pronounced at $\gamma_M = 45$ and 90 degrees. Thus, the alignment of the magnetic field plays a pivotal role in shaping the flow dynamics and optimizing the transfer processes, adding a fascinating layer to the intricate interplay of forces in the system. As a result, for effectively controlling convection phenomena, the magnetic field angle can be strategically utilized, holding particular significance in modern semiconductor device manufacturing for crystal growth.

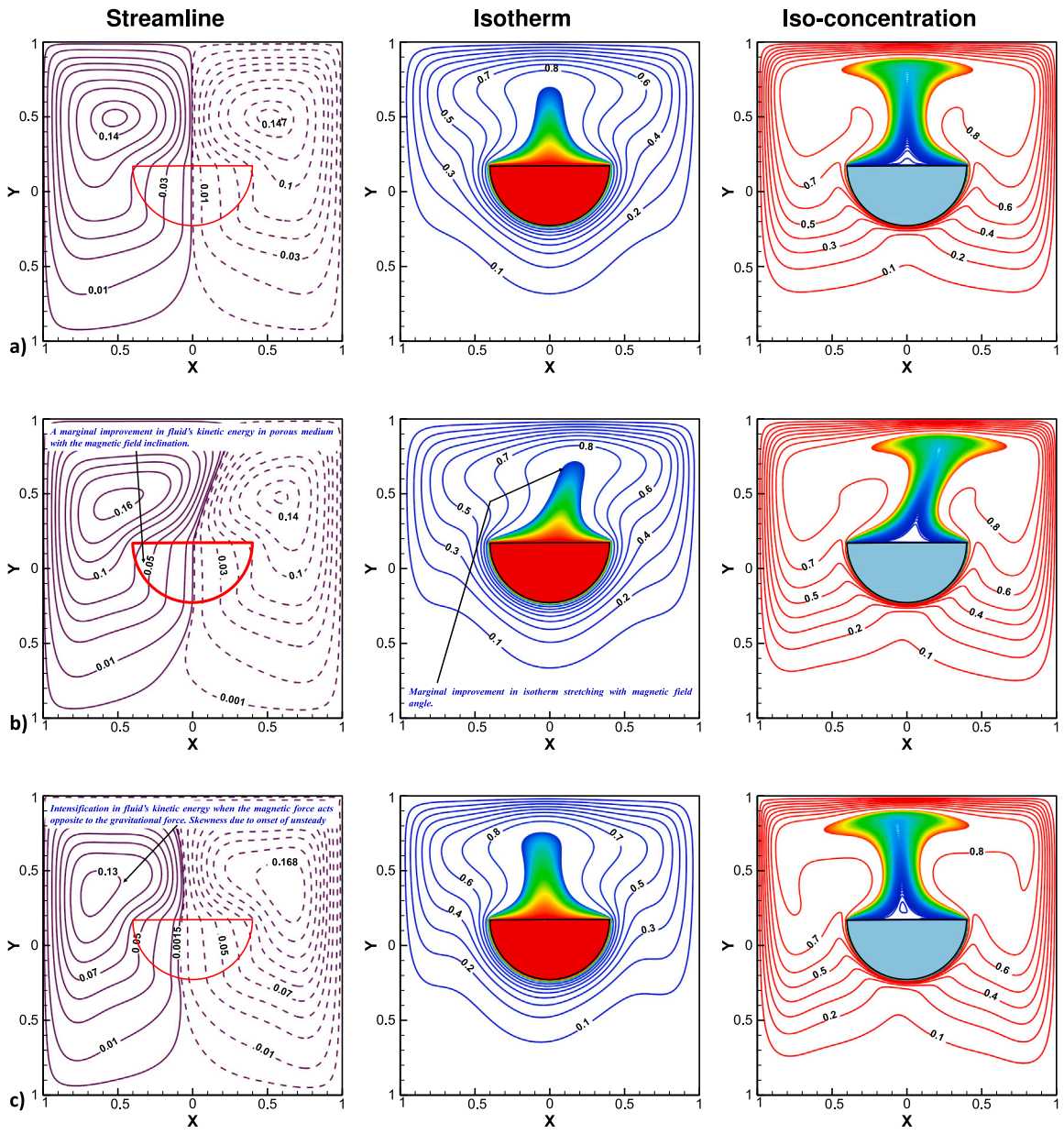


Fig. 6. Significance of the magnetic field angle (γ_M) on the configuration of streamlines, isotherms and iso-concentration lines under the conditions of $Ha = 25$, $BR = 2$, and $Da = 10^{-3}$, displayed for (a) $\gamma_M = 0°$, (b) $\gamma_M = 45°$, and (c) $\gamma_M = 90°$.

4.1.5. Dynamic nature of heat and concentration transport

The interplay of diverse elements investigated in this investigation gives rise to time-varying or repeating patterns in the flow, heat, and concentration distributions. To comprehend the impact of these elements, the frequency spectra of the average Nusselt number (Nu_M) and Sherwood number (Sh_M) are plotted in Figs. 7(a)–(f) for various Da and γ_M values, considering $BR = -5$ and $Ha = 25$. Observations reveal that altering the parameters initiates periodic occurrences. As the fluid's kinetic energy gains more momentum, it progresses to transient or oscillating behaviour in heat and concentration attributes. This unsteady pattern of change is a consistent trend witnessed in the dynamic evolution across the various parameters explored in this investigation. In a general sense, when a parameter substantially boosts the fluid's kinetic energy, an unsteady pattern comes into play. For instance, the amplification of γ_M and Da elevates the fluid's kinetic energy, leading to a noticeable increase in the amplitudes of Nu_M. Nevertheless, when considering Sh_M, the rise in permeability brings about a decrease in the amplitude of Sh_M, while simultaneously enhancing the frequency of its peaks. This effect is particularly prominent when $\gamma_M = 45°$ and $90°$. Notably, the most pronounced peak of Sh_M is observed at $\gamma_M = 90°$ when $Da = 10^{-4}$. Conversely, the frequency of unsteadiness of this Da is lesser than that of higher Da value. Hence, it is evident that either the amplitude or the frequency of oscillation is enhancing with the augmented kinetic energy offered by the Da and γ_M. The predominant frequency of Sh_M and Nu_M is observed when $\gamma_M = 90°$ than other orientations. As mentioned earlier, the transfer of concentration outpaces heat transfer, contributing to a slightly higher amplitude of unsteady behaviour in Sh_M in comparison to Nu_M. Furthermore, the larger flow volume traversing the porous zone due to the increased Da values magnifies the thermal and concentration gradients. This, in turn, strengthens the buoyant force and enhances the fluid's kinetic energy within and around the porous semi-circular structure. On the other hand, the manipulation of Ha and BR can wield considerable influence in shaping the unsteady pattern under the influence of γ_M and Da. The flow dynamics resulting from the interplay of these various parameters are visually depicted in Fig. 8. Generally, unsteady features become

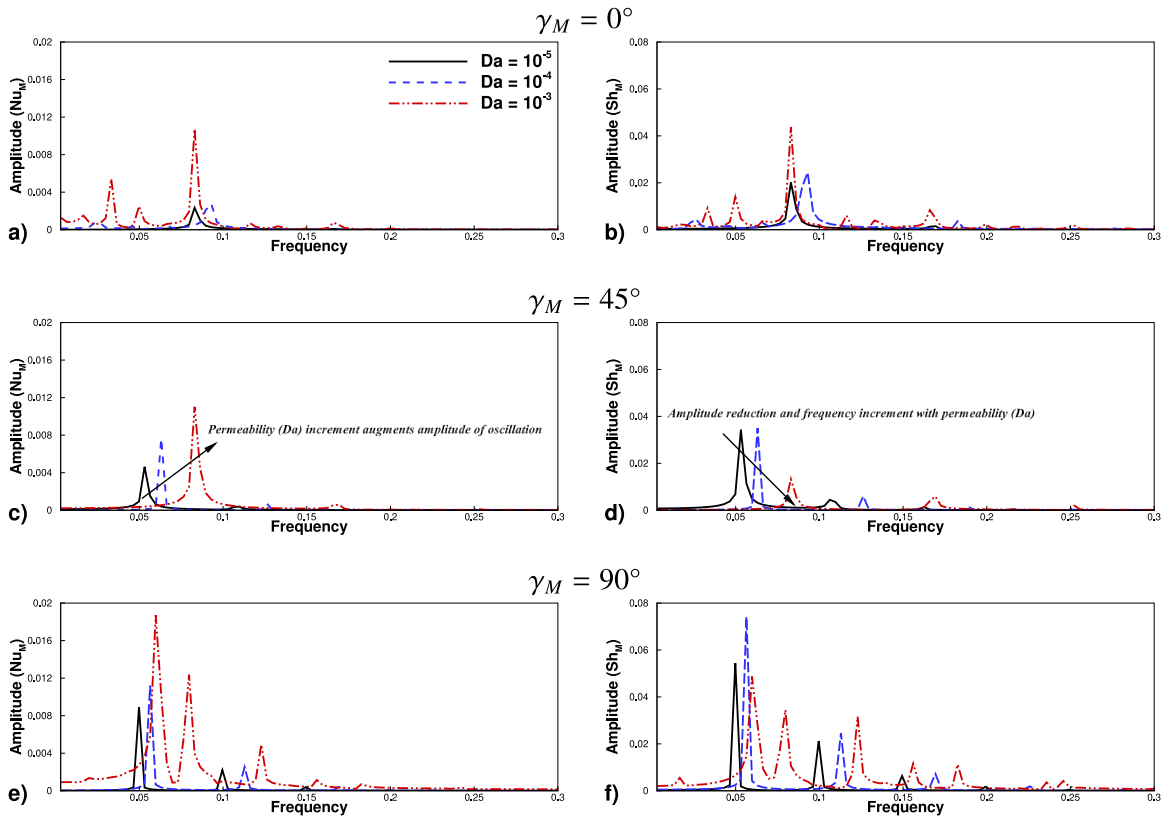


Fig. 7. Frequency spectra of average Nusselt number (Nu_M), and average Sherwood number (Sh_M) across various magnetic field angle (γ_M) and Darcy number (Da) under the conditions of $Ha = 50$ and $BR = -5$.

noticeable when BR assumes negative values. This can be attributed to the counterflow direction exhibited by the fluid at such BR values, which drives the fluid towards the flat edge of the semi-circular body, causing early separation of the fluid boundary layer. Simultaneously, the thermal buoyant force counteracts the concentration buoyant force, adding to the unsteady nature of the flow. These factors play a significant role in establishing the predominance of unsteady features at negative BR values. Moreover, this phenomenon can experience a complete transition into a steady state or undergo a decrease in the amplitude of the unsteady pattern as the horizontal intensity of the magnetic field increases (i.e. $\gamma_M = 0°$), as depicted in Fig. 8 for the scenario when $Ha = 50$ and $\gamma_M = 0°$. Setting the magnetic angle to 45° triggers the emergence of periodic or oscillatory attributes in heat and concentration transfer behaviours, even when $Ha = 50$. Moreover, unsteady characteristics are also observed at positive BR values, especially when $\gamma_M > 0°$. The interaction between an aligned magnetic field and the localized amplification of kinetic energy near the curved fore-body, particularly at higher BR values, accounts for these unsteady patterns. The oscillating pattern in concentration and heat transfer becomes evident at positive BR values, particularly when $Ha = 50$ and $\gamma_M = 45°$ or 90°. This phenomenon can be attributed to the inclined flow of the fluid caused by γ_M, combined with the heightened kinetic energy observed at higher positive BR values. Furthermore, the magnetic field angle wields a more pronounced effect when Ha is elevated. As a result, the periodic unsteady behaviours become more conspicuous at $Ha = 50$, in contrast to the scenario with $Ha = 25$, particularly when BR takes positive values. In a broader sense, the interplay between the flat and curved surfaces of the permeable body, in conjunction with the magnetic field and its orientation, plays a pivotal role in shaping the flow dynamics.

4.1.6. Mean Nusselt number (Nu_M)

The mean Nusselt number (Nu_M) serves as the key determinant in evaluating the dissipation of heat to the cavity walls. When considering transient scenarios, the values of Nu_M and Sh_M are subject to temporal averaging. The figures presented in Figs. 9(a)–(c) illustrate the variations of Nu_M in relation to the parameter BR, across a range of Ha, Da, and γ_M values. Prior to delving into the distinctive findings concerning the interplay of different parameters on Nu_M as observed in this investigation, it would be beneficial to initially grasp the traditional or classical outcomes linked with each individual parameter. At $BR = -1$ an equilibrium emerges, wherein the buoyant forces originating from concentration and temperature intricately counteract one another. This balance significantly impedes the predominant impetus driving fluid dynamics, culminating in marginal heat transfer specifically at this BR value. As the negative values of BR progress further, the potency of the buoyant force arising from concentration experiences augmentation, subsequently leading to an amplification in heat transfer rates. Nevertheless, as permeability advances, the alterations in Nu_M in relation to changes in BR become more conspicuous, showcasing more pronounced and abrupt transformations. The influence of Da exhibits its least substantial effect when BR attains -1, while conversely, it demonstrates its most conspicuous impact at the value of 5, regardless of the prevailing magnitudes of the Ha and γ_M. The act of elevating both Da and γ_M serves to heighten the Nu_M, thereby instigating an augmentation in the efficiency of heat transfer. Moreover, an increase in the Ha serves to impede the performance of thermal dissipation. The augmentation in the Nu_M value attributed to variations in Da manifests a noticeable enhancement during the transition from $\gamma_M = 0°$ to $\gamma_M = 45°$, whereas at 90°, this enhancement experiences a marginal diminishment. As an illustration, consider the scenario where $Da = 10^{-3}$ (compared to 10^{-5}), $Ha = 50$, and $BR = 1$, the percentage elevation in Nu_M is quantified as 15.49% for $\gamma_M = 0°$, 26.79% for $\gamma_M = 45°$, and 21.71% for $\gamma_M = 90°$. Remarkably, the most substantial augmentation, reaching its peak, is observed for $\gamma_M = 45°$, surpassing all other magnetic force orientations for the entire spectrum of BR levels. Furthermore, the enhancement conferred by the shift

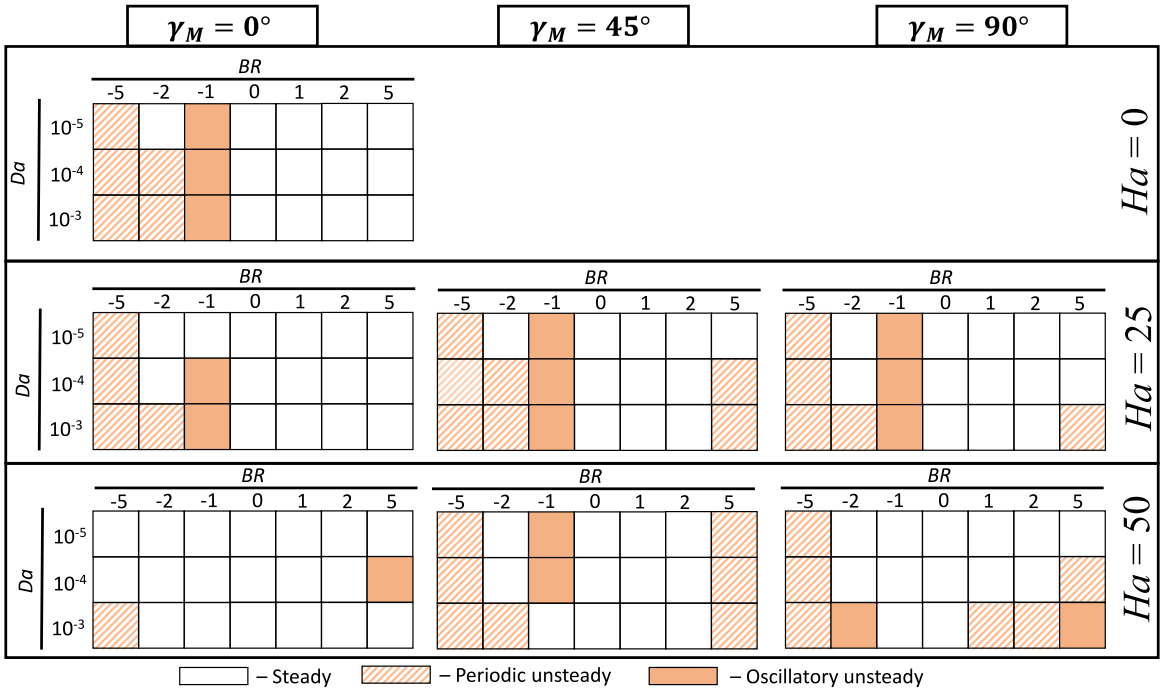


Fig. 8. Variation of flow regime (steady & unsteady) across the diverse factors investigated in the analysis.

from $Da = 10^{-5}$ to 10^{-3} registers a significantly more pronounced magnitude when compared with the scenario involving $Da = 10^{-4}$. Moreover, the decline in Nu_M induced by an elevation in the Ha exhibits a diminishing trend as the γ_M experiences an increase. To elucidate, consider the case of $Ha = 50$, $Da = 10^{-3}$, and $BR = 1$. In this scenario, the percentage decrement in Nu_M is quantified as 43.15% for $\gamma_M = 0°$, 33.72% for $\gamma_M = 45°$, and 26.06% for $\gamma_M = 90°$ in comparison to $Ha = 0$. Notably, as γ_M reaches 90°, the influence of the magnetic field in impeding heat transfer exhibits a pronounced diminishment. Undoubtedly, these revelations underscore the potential to finely regulate the magnetic field's impact by adeptly manipulating the angle of the magnetic field. Furthermore, aligning the magnetic field in opposition to gravity and introducing heightened permeability stands as a strategy to fortify thermal efficiency. Besides, the anomalous peaks witnessed in Nu_M are discernible exclusively under the influence of the magnetic field. Across all such instances, the augmentation in Nu_M is ascribed to the emergence of a singular robust vortex as opposed to the presence of dual vortices. As shown in Fig. 9(a), the mean Nusselt numbers achieve their pinnacle at $BR = -1$ for $Ha = 25$, and at $BR = -2$ for $Ha = 50$, when $Da = 10^{-5}$. The corresponding thermal distribution depicted within the plot illustrates the formation of a counter-clockwise vortex and a clockwise vortex for the scenarios of $BR = -1$ and $BR = -2$, respectively. As a result, a greater expanse of the permeable semi-circular surface becomes exposed to the colder fluid, consequently yielding a heightened pace of thermal dissipation. With the amplification of magnetic field strength, the prevalence of x-direction velocity gains ascendancy, leading to the lateral elongation of the single vortex formation particularly at $Ha = 50$. The escalation in permeability provokes the inception of a uni-cell vortex in situations where the buoyant force arising from concentration is non-existent. Consequently, at $Da = 10^{-4}$, an observable upswing in Nu_M becomes evident when $BR = 0$. Furthermore, the atypical outcomes of heat transfer outcomes are absent when $\gamma_M = 45°$. Nonetheless, at $\gamma_M = 90°$, under the conditions of $Da = 10^{-3}$ and $Ha = 25$, an unsteady unicellular vortex with a reduced amplitude takes form when $BR = -5$. The amalgamation of flat and curved boundaries within the permeable structure, coupled with the strength of the magnetic field, converges to engender the emergence of this singular vortex, precisely when the fluid's kinetic energy attains an optimal magnitude. Delving into the mechanisms underpinning the inception of this unicellular vortex stands as a captivating avenue for further exploration and inquiry.

4.1.7. Mean sherwood number (Sh_M)

The evolution of the enclosure's Sherwood number (Sh_M), influenced by variations in BR, Ha, Da, and γ_M, is visually depicted in Figs. 10(a)–(c). Particularly noteworthy is that when the Le is assigned a value of 5, the concentration's boundary layer evinces a reduction in thickness as compared to the thermal boundary layer. As a result, the corresponding Sh_M values manifest as relatively higher than their Nu_M counterparts, regardless of alterations in other parameters. Additionally, the trajectory of Sh_M in relation to BR follows a parallel pattern to that observed in Nu_M concerning its response to changes in BR. Furthermore, both Da and γ_M exert a positive impact on the Sh_M, while Ha impedes the process of concentration transfer. Specifically, when $BR = -1$, the influence of Da, Ha, and γ_M on Sh_M remains relatively limited. However, as BR assumes a value of 5, these factors emerge as pivotal determinants in shaping the magnitude of Sh_M. The escalation in Sh_M induced by the augmentation of permeability becomes progressively more pronounced with the increase in γ_M, particularly at elevated levels of BR. Illustratively, consider the case where $Da = 10^{-3}$ (in comparison to 10^{-5}), $Ha = 50$, and $BR = 1$. In this context, the percentage augmentation in Sh_M is quantified as 56.07% for $\gamma_M = 0°$, 69.68% for $\gamma_M = 45°$, and 55.99% for $\gamma_M = 90°$. Analogously to the observation in the Nu_M scenario, the most pronounced enhancement becomes evident when γ_M attains 90°. Notably, the influence of permeability on Sh_M exhibits a distinctly more accentuated character in comparison to its impact on Nu_M, as portrayed in Section 4.1.6. Moreover, the augmentation witnessed in Sh_M is more pronounced upon elevating Da from 10^{-4} to 10^{-3}, in contrast to the enhancement observed from 10^{-5} to 10^{-4}. Conversely, γ_M demonstrates its propensity to ameliorate the decline in concentration transfer that arises due to an increment in Ha. To exemplify, when comparing $Ha = 50$ with $Ha = 0$, both under the conditions of $Da = 10^{-3}$ and $BR = 1$, the percentage reduction in Sh_M amounts

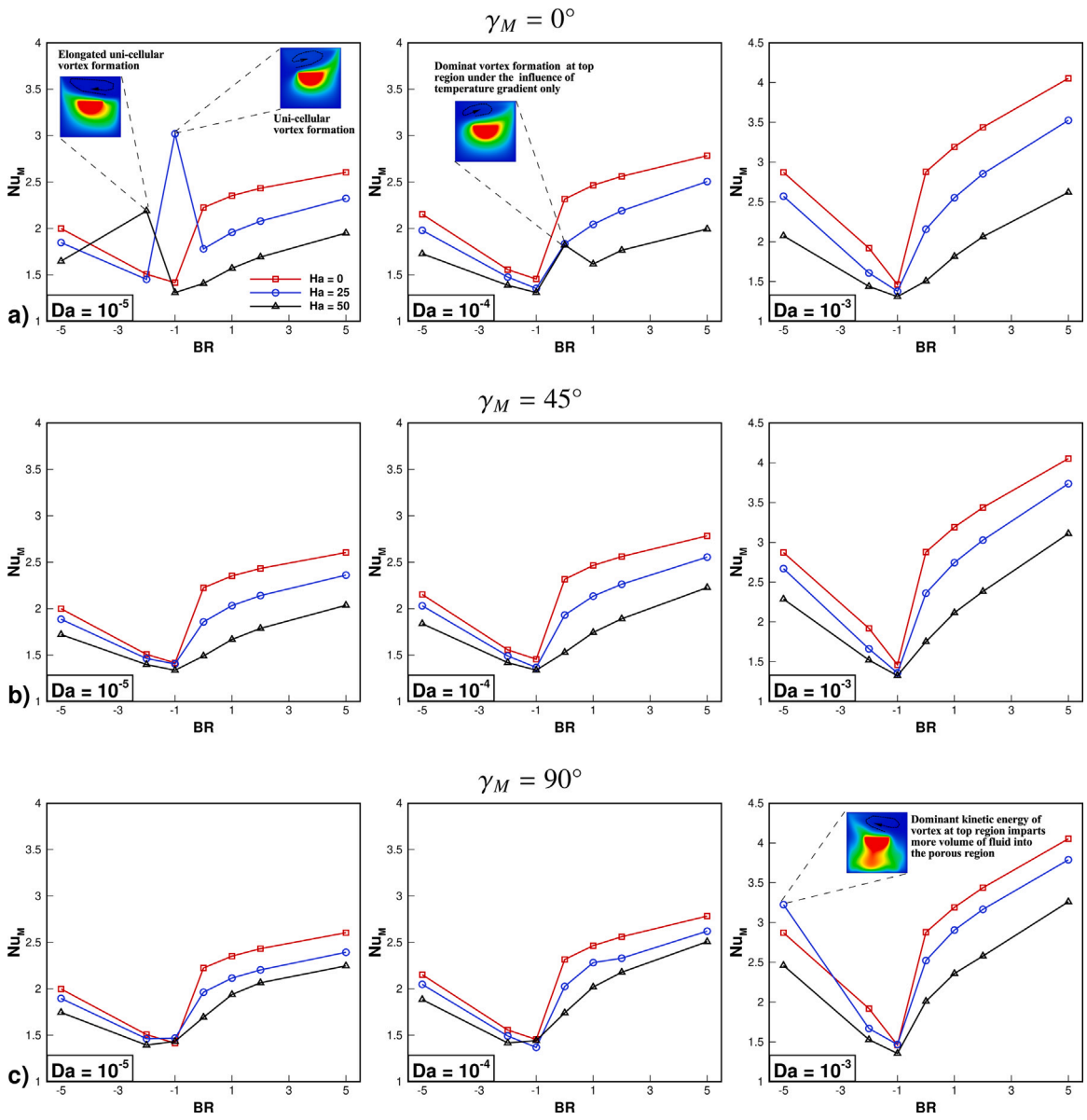


Fig. 9. Fluctuations in mean Nusselt number (Nu_M) of the cavity across distinct buoyancy ratio (BR), under different conditions of Darcy number (Da) and Hartmann number (Ha) at (a) $\gamma_M = 0°$, (b) $\gamma_M = 45°$, and (c) $\gamma_M = 90°$.

to 38.12% for γ_M at 0°, 26.17% for γ_M at 45°, and 21.99% for γ_M at 90°. Notably, the pronounced inhibition exerted by the magnetic effect on concentration transfer undergoes a substantial attenuation when the magnetic field angle γ_M is maintained at 90°. Moreover, in a manner akin to the observations made for Nu_M, the emergence of a solitary vortex formation is also witnessed in the exceptional augmentation of Sh_M. However, it is worth highlighting that the parameter combinations giving rise to this characteristic behaviour in Sh_M do not align consistently with those observed in Nu_M. For instance, although the occurrence of a single vortex formation becomes evident at $Ha = 50$, $Da = 10^{-5}$, and $BR = 5$, no corresponding surge in Sh_M is observed. It is worth reiterating that, within the framework of these specific parameter combinations, an atypical peak in Nu_M was observed. This phenomenon can be attributed to an escalated convective flow rate, leading to the formation of substantial boundary layers on both sides of the curved surface. As a consequence, the fluid emanating from these regions induces a reduction in the concentration gradient, which in turn suppresses the occurrence of the unconventional behaviour in Sh_M.

5. Concluding remarks

A numerical exploration was executed to scrutinize the profound impact of several factors on the captivating dynamics of double-diffusive natural convection within a container housing a concentrated and heated semi-circular permeable structure. The primary focal point of this inquiry resided in the Darcy number (Da) with three selected values: 10^{-5}, 10^{-4}, and 10^{-3}, the buoyancy ratio (BR) spanning from −5 to +5, the Hartmann number (Ha) ranging from 0 to 50, and the magnetic field angle (γ_M) encompassing 0°, 45°, and 90°. The Lattice Boltzmann method was employed to carry out this investigation. Strengthening the magnetic effect impedes the gradients of concentration and temperature around the permeable structure, resulting in a reduction of Nu_M and Sh_M with the escalation of the Ha. Nevertheless, despite the substantial influence on the semi-circular body, this augmentation only results in a marginal alteration in the flow field's intensity within the permeable zone. Aligning the magnetic force with the direction of flow (*i.e.* $\gamma_M = 90°$) heightens the kinetic energy, culminating in heightened heat and concentration attributes. A rise in

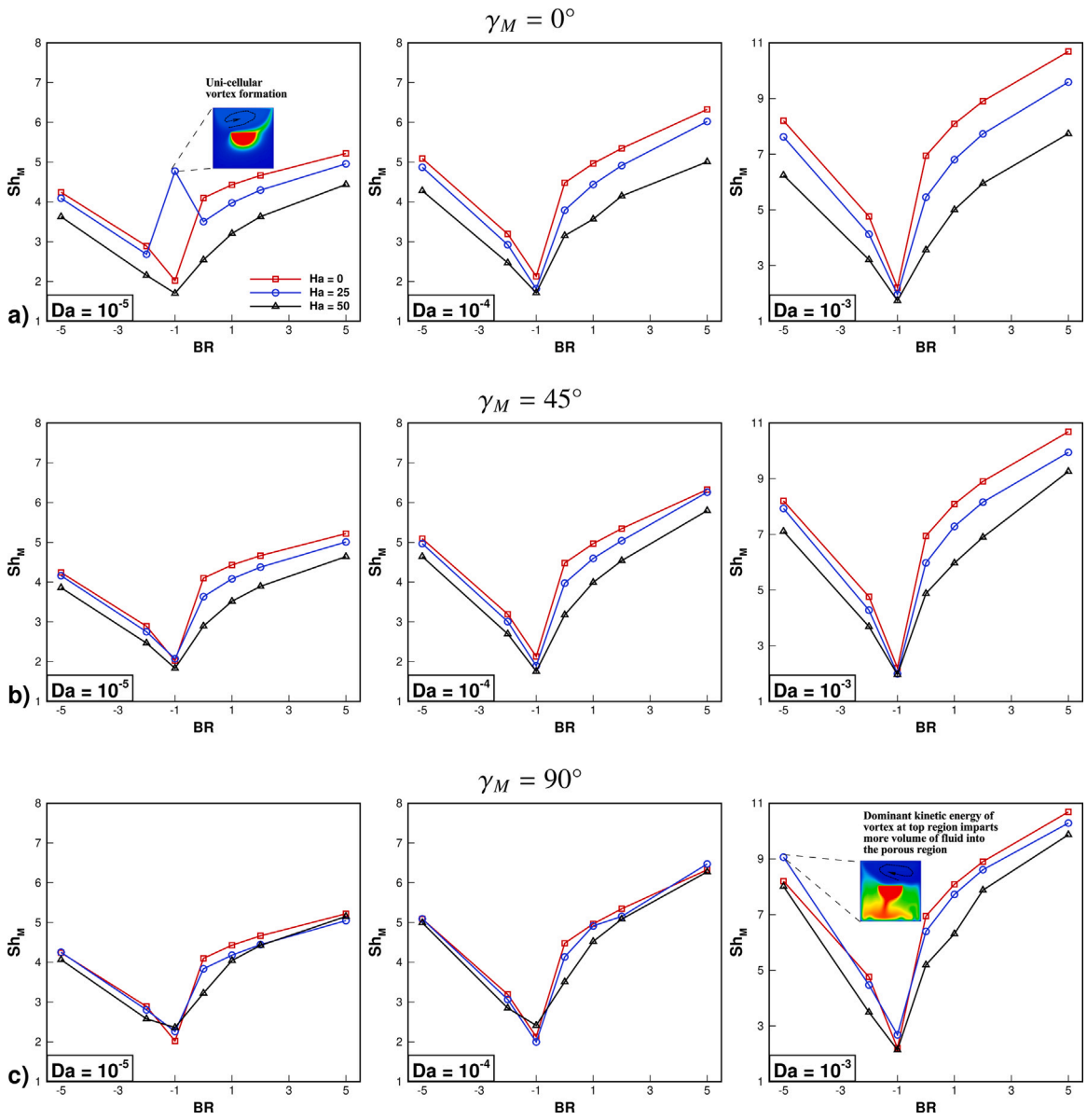


Fig. 10. Fluctuations in mean Sherwood number (Sh_M) of the cavity across distinct buoyancy ratio (BR), under different conditions of Darcy number (Da) and Hartmann number (Ha) displayed for (a) $\gamma_M = 0°$, (b) $\gamma_M = 45°$, and (c) $\gamma_M = 90°$.

permeability results in a substantial increase in heat and concentration rates. As the Da increases for the semi-circular body, it further accentuates the rise in localized thermal and concentration gradients. This, in turn, leads to an amplification of the fluid's kinetic energy in the vicinity of the semi-circular region. Additionally, an increase in the BR propels a greater volume of fluid to advance through the permeable semi-circular body with heightened intensity, consequently ensuring the augmentation of heat and concentration traits. As the magnetic field angle (γ_M) rises, fascinating periodic and oscillatory flow phenomena come to the forefront. The amplitude of these oscillations grows as both the Da and γ_M increase. Notably, the dominance of negative BR becomes evident in the emergence of these unsteady characteristics, particularly in the presence of magnetic field. Moreover, the increase in γ_M diminishes the influence of Ha on the reduction of Nu_M and Sh_M, while concurrently heightening the effect of Da on the increase of Nu_M and Sh_M. In specific configurations of parameters, a sudden surge in heat and concentration transfers becomes evident. This phenomenon arises due to the formation of a robust single vortex at the upper region of the cavity. The intricate interplay among flat and curved surfaces, permeability, and the magnetic field gives rise to this anomalous pattern in Nu_M and Sh_M. Venturing into the flow, heat, and concentration dynamics unveils the substantial impacts of permeability, buoyancy ratio, magnetic field, and its orientation on control mechanisms. Understanding a permeable semi-circular shape is crucial for figuring out unusual flow patterns. Going deeper into the details would enable us to use this special shape in heating devices, opening up new engineering possibilities.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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